

Review

Deployable image forgery detection and localization: A comprehensive review and practitioner's guide

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ABSTRACT

Digital images underpin reporting, science, security, and social communication, yet modern editing and generative tools make manipulation effortless and often imperceptible. Digital image forgery detection is fundamentally a pattern recognition problem: the task is to learn and exploit stable patterns in authentic imagery and to recognize distributional departures introduced by manipulations. This review synthesizes research on image forgery detection and localization (IFDL) from 2016 to 2025, spanning traditional passive forensics, CNN-based detectors, and the latest transformer-driven models. We formalize a two-axis taxonomy, using a technical approach (handcrafted / statistical, CNNs, transformers, GAN- and multimodal-assisted) and by forgery type (copy-move, splicing, object removal / inpainting, retouching, and AI-generated fakes). Classical methods exploit lighting, compression, noise, and block / keypoint matching for copy-move and splicing; deep models advance to pixel-wise localization with multi-stream cues (RGB, noise / frequency, edges) and robust pretraining. We analyze publicly used datasets (e.g. Columbia, CASIA, MICC, CoMoFoD, COVERAGE, NIST16, Realistic Tampering, PS-Battles, IMD2020, DEFACTO, tampCOCO, DF2023), highlighting scale, annotation granularity, and manipulation diversity, and relate them to reported performance trends. A critical gap analysis identifies persistent challenges: cross-dataset generalization, robustness to laundering (compression, blur, resizing), explainability and evidentiary utility, data scarcity for subtle edits, and efficiency for large-scale deployment.

1. Introduction

Digital images have become an important source of information in journalism, law, medicine, military intelligence, and social networks. At the same time, advances in image editing software have made it trivially easy to manipulate images. As a result, determining whether a digital image has been tampered with is now a critical challenge. Image forgery can range from simple enhancements (retouching) to creating completely fabricated scenes (splicing or deepfakes). For example, image retouching only subtly alters the appearance of an image (e.g., adjusting colors or sharpness) without changing its content. In contrast, image splicing combines content from multiple images to produce a fake composite. Another common form is copy-move forgery, where a region from the image itself is copied and pasted elsewhere in the same image, often to duplicate or conceal an object. These tampering operations are usually imperceptible to the naked eye, which motivates the need for automated detection.

Digital image forensic techniques are broadly classified into active and passive approaches. Active methods (e.g. digital watermarking or signatures) embed a secret code into the image at the time of creation, so any alteration can later be detected by the absence or corruption of this

code. However, most consumer cameras do not apply watermarks or signatures. Therefore, real-world forgery detection usually relies on passive (or blind) techniques, which do not assume any pre-embedded signals. Passive image forensics works by analyzing the inherent statistics and inconsistencies of the image to identify tampering (Farhan et al., 2024). Different types of forgery leave different traces; thus, a variety of specialized detectors have been developed. For example, splicing might be revealed by lighting inconsistencies between composited objects or by discrepancies in camera-specific artifacts like color filter array patterns. Copy-move forgeries can be caught by detecting duplicated regions via pattern matching. No matter the method, the overarching goal is to verify the integrity and authenticity of digital images, especially when they are used as evidence or in high-stakes domains.

1.1. Research motivation

The prevalence of image manipulation poses a serious threat to the reliability of visual media. Forged images have been used to spread misinformation ("fake news"), involve people in crimes, commit insurance fraud, and even scientific evidence. In particular, a study found that 3.8% of a sample of scientific papers contained manipulated images,

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which cost an estimated \$392k per retracted paper (Bik et al., 2016). Such incidents highlight the urgent need for robust forgery detection. Rapid progress in deepfake technology since 2017 has further exacerbated the problem. Today's forgers can employ advanced generative models to produce highly realistic fake images and videos, making manual detection nearly impossible. This raises key research questions: How can we automatically detect and localize manipulated regions in an image? What techniques are effective against modern AI-assisted forgeries? How can detection methods keep pace with increasingly realistic fakes?

1.2. Objective and scope

This review paper aims to comprehensively survey the state of the art in image forgery detection and localization from 2016 through 2025. We cover both traditional digital image forensic techniques and modern machine learning approaches, with an emphasis on recent deep learning-based methods (convolutional neural networks (CNN) and vision transformers (ViT)). The focus is on passive forgery detection in single images. We address detection of common forgery types: copy-move, splicing, object removal (inpainting), and cosmetic retouching, as well as emerging categories such as generative adversarial network (GAN)-generated deepfakes. For each category, we discuss representative algorithms, their underlying techniques, and performance on benchmark datasets. Our objective is to highlight the evolution of this field, from early methods to current AI-driven techniques, and to identify the strengths and limitations of each approach. We also aim to define a taxonomy of methods to organize the literature and to pinpoint knowledge gaps and challenges that present opportunities for future research.

1.3. Contribution to knowledge

By consolidating more than a decade of research, this paper provides a timely overview of image forgery detection techniques. Unlike previous surveys that covered either traditional approaches or early deep learning methods, our review bridges both eras and extends to the latest advances in transformers and generative models. We critically analyze how the field has progressed, e.g., the shift from manual feature engineering to data-driven feature learning, and how well current techniques address real-world forensic needs. A detailed gap analysis is presented, discussing issues of generalization, robustness, and evaluation that per-

sist despite technological progress. Finally, we propose future directions and promising research avenues to advance image integrity verification.

Table 1 frames the review around three guiding research questions and their rationale, which collectively structure the rest of the articles. *RQ1* motivates a unified taxonomy that spans traditional, CNN-based, and transformer / fusion approaches and explains why mapping techniques to manipulation families (copy-move, splicing, inpainting, retouching, AI-generated) is essential to synthesize a decade of progress. *RQ2* centers on empirical validity, how methods perform across datasets, metrics, and real-world degradations, and thus justifies our emphasis on dataset coverage, annotation granularity, and robustness testing. *RQ3* foregrounds open problems (generalization, counter-forensics, precise localization, and explainability) and provides a roadmap for future work. Taxonomy and data set atlas (Sections 3 and 4 address *RQ1*, method analyzes (Sections 5.1, 5.2 and 5.3) and robustness evidence address *RQ2*, and the discussion plus research agenda (Sections 6 and 7) synthesize answers to *RQ3* and define priorities to advise trustworthy image integrity verification.

The theoretical contributions of our work are as follow:

- *Unified IFDL taxonomy*: We develop a two-axis taxonomy of image forgery detection and localization methods that jointly organizes techniques by underlying paradigm (traditional, CNN-based, transformer-based) and by manipulation type (copy-move, splicing, inpainting), extending earlier, single-axis taxonomies and clarifying how recent deep models relate to classical approaches (Sections 2 and 3).
- *Evidence map of methods, datasets, and metrics*: We construct a structured mapping of articles primary studies to their tasks, annotation granularity, datasets, and evaluation metrics, providing a consolidated view of how IFDL research is actually evaluated in practice (Section 4).
- *Cross-cutting gap analysis*: By comparing results across architectures and forgery families, we analyze open problems in generalization, robustness to laundering, and explainability, highlighting where current detectors fail and where existing benchmarks are insufficient (Sections 5 and 6).

The practical contributions of our work are summarized as follows:

- *Recommendations on datasets and evaluation protocols*: We provide concrete suggestions on selecting public datasets, designing

Table 1
Research questions addressed in this review and their motivations.

Research Question (RQ)	Motivation
<i>RQ1: What are the main categories of approaches for image forgery detection and localization from 2016 to 2025, and how have these techniques evolved over this period?</i>	The field of image forensics has grown rapidly with the introduction of deep learning. This question aims to categorize the diverse methods (from traditional to CNN-based to transformer-based) and to map out the evolution of approaches. By doing so, we can understand how the state of the art arrived at its current point, highlighting the progression of ideas, the incorporation of new technologies, and the paradigm shifts. This structured overview provides a foundation for researchers to identify which families of techniques have been most influential and where new contributions could be made.
<i>RQ2: How do current forgery detection methods perform on different types of image manipulations (e.g., copy-move, splicing, inpainting, GAN-generated deepfakes), and what are their evaluation benchmarks and limitations?</i>	Forgery detection is not a one-size-fits-all problem; performance can vary greatly depending on the type of manipulation and conditions like image quality. <i>RQ2</i> is motivated by the need to assess the effectiveness of existing methods in various scenarios. By comparing results across multiple benchmark datasets and forgery categories, we can pinpoint strengths and weaknesses. This question also encourages a critical look at common evaluation practices, helping to identify gaps, for example, forgery types or real-world conditions that have not been sufficiently covered in testing.
<i>RQ3: What are the major open challenges in image forgery detection and localization, and what future research directions emerge from these challenges?</i>	Despite numerous advances, image forgery detection remains an arms race against increasingly realistic and complex fakes. This question is driven by the need to illuminate the road ahead. By enumerating the open challenges, like generalizing to novel forgeries, countering anti-forensic measures, improving localization precision, and making models explainable, we set the stage for future innovation. The motivation is to ensure that the research community is aware of the critical pain points that must be addressed to move from today's promising results to robust, real-world forensic solutions. Discussing future directions linked to these challenges can inspire targeted research that pushes the field closer to those goals.

train/test and cross-dataset splits, and reporting both image-level and pixel-level metrics to support reproducible, deployment-grade evaluation of IFDL systems (Sections 4 and 6).

- **Model-selection guidance for deployment:** We distill design guidelines that relate detector families (traditional, CNN, transformer) to operational constraints such as latency, throughput, and hardware budget, helping practitioners choose suitable models for triage, localization, or high-fidelity audit scenarios (Section 6).

1.4. Paper structure

The remainder of the paper is organized as follows. [Section 1](#) motivates the problem, differentiates active vs. passive forensics, and states the objective and scope of this review. [Section 2](#) concerns the background, summarizing the evolution of image forensics, and positions recent deep and transformer trends. [Section 3](#) describes the literature search and selection process and presents the hierarchical taxonomy that organizes approaches by evidence type and class of manipulation. [Section 4](#) compiles the major benchmarks for detection / localization, noting manipulation coverage, annotation granularity, and typical evaluation splits. [Section 5.1](#) reviews copy-move forgery detection (CMFD) block / keypoint / dense field and splicing detectors grounded in signal processing and classical machine learning (ML), with strengths / limitations. [Section 5.2](#) surveys CNN single- and multi-stream for detection and pixel-level localization, including edge / noise / frequency fusion and multitask heads. [Section 5.3](#) covers ViT / Swin and fusion transformers that capture long-range context and integrate red green blue (RGB) with noise / frequency evidence. [Section 6](#) discusses the answers to the three research questions, compares methods between forgery types and conditions, analyzes robustness gaps, and distills design takeaways. Finally, conclusive remarks and perspective work are drawn in [Section 7](#), summarizing the contributions and outlining the research agenda for generalizable, explainable and deployment-grade forensics.

2. Research background

Digital images have evolved into a primary vehicle for information exchange in journalism, social media, medicine, law enforcement, and

scientific publishing. Their ubiquity, combined with the rapid democratization of sophisticated editing tools and artificial intelligence (AI)-based content generators, has sharply increased the prevalence and realism of forgeries, from traditional copy-move and splicing manipulations to today's GAN-synthesized deepfakes. Early forensic research focused on handcrafted statistical cues such as camera sensor noise, chromatic aberration, and joint photographic experts group (JPEG) re-compression artifacts to flag tampered regions. Although these techniques laid essential theoretical foundations, their reliance on narrowly tuned features limited robustness and scalability. The emergence of deep learning around 2016 triggered a paradigm shift: convolutional neural networks (CNNs) began to learn manipulation traces directly from data, dramatically boosting detection accuracy, while fully convolutional variants enabled pixel-level localization of falsified content. Most recently, vision transformers and hybrid CNN-Transformer architectures have further advanced the field by capturing long-range contextual inconsistencies and fusing multi-modal evidence. Despite these gains, challenges such as cross-dataset generalization, adversarial robustness, and interpretable decision-making remain open, motivating ongoing research to develop more resilient, explainable, and universally applicable forgery detection frameworks (Khan et al., 2025).

[Table 2](#) presents a comprehensive comparison of selected review articles published from 2016 to 2025, while [Fig. 2](#) highlights the number of review articles published in the same time period, classified according to the types of image-forgery techniques and the models they discuss. The forgery types considered include copy-move, image splicing, deepfake, inpainting, morphing, and GAN-based forgeries. Each reference is examined to determine its coverage of these forgery classes. Furthermore, the papers are classified by the types of model on which they focus, such as traditional hand-crafted methods, conventional machine learning algorithms, convolutional neural networks (CNNs), deep learning architectures, and transformer-based models. This classification enables the reader to identify research trends and gaps between different forgery categories and modeling techniques. Additionally, the dataset column highlights whether the review explicitly discussed publicly available datasets used for benchmarking. This tabular representation facilitates a systematic understanding of the scope and contributions of existing review articles in the domain of image forgery detection and segmentation. Each reference is examined to determine its coverage of these

Table 2

Recent reviews on image forgery: covered forgery types, model categories, and dataset usage.

Ref	Forgery Type						Model Type					Datasets
	CM	Splice	DF	Inpaint	Morph	GAN	Trad.	ML	CNN	DL	Trans.	
Dehghani and Saberi (2025)	×	×	✓	×	✓	✓	✓	✓	✓	✓	✓	×
Singh et al. (2025a)	×	×	✓	×	×	✓	×	✓	✓	✓	×	×
Kumari and Garg (2025)	✓	✓	×	×	×	×	✓	✓	✓	✓	×	✓
Kaur et al. (2024)	✓	✓	×	✓	×	×	✓	✓	✓	✓	×	×
Muthaiah et al. (2024)	×	×	✓	×	✓	✓	×	×	✓	×	✓	✓
Passos et al. (2024)	×	×	✓	×	✓	✓	×	✓	✓	×	✓	✓
Singh and Kumar (2024)	✓	✓	×	✓	✓	✓	✓	✓	✓	✓	×	✓
Barglazan et al. (2024)	✓	✓	×	✓	×	✓	✓	✓	✓	✓	✓	✓
Edwards et al. (2024)	×	×	✓	×	×	✓	×	✓	✓	×	✓	✓
Heidari et al. (2024)	×	×	✓	×	✓	✓	×	✓	✓	✓	×	×
Gupta et al. (2023)	×	×	✓	×	×	✓	✓	✓	✓	✓	×	✓
Pandey et al. (2023)	✓	✓	×	✓	×	×	×	×	×	✓	×	✓
Shi et al. (2023)	✓	✓	×	✓	×	✓	✓	×	✓	✓	×	✓
Kaur et al. (2023)	×	×	×	×	×	×	✓	✓	✓	×	×	✓
Castillo Camacho and Wang (2021)	✓	✓	✓	✓	×	×	✓	✓	✓	×	×	✓
Ferreira et al. (2020)	✓	×	×	×	×	×	✓	×	×	✓	×	×
Thakur and Rohilla (2020)	✓	✓	×	×	✓	✓	✓	×	✓	×	×	✓
Rani and Kumar (2019)	✓	✓	×	✓	×	×	✓	×	×	×	×	×
Teerakanok and Uehara (2019)	✓	×	×	×	×	×	✓	✓	✓	×	×	×
Dixit and Gupta (2016)	✓	✓	×	×	×	×	✓	×	×	×	×	×
Our Work	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

CM = Copy-move, Splice = Image-splicing, DF = Deepfake, Inpaint = Image-inpainting, Morph = Morphing, GAN = Generative adversarial network, Trad. = Traditional model, ML = Machine learning, CNN = Convolutional neural network, DL = Deep learning. ✓ = addressed / used; × = not addressed.

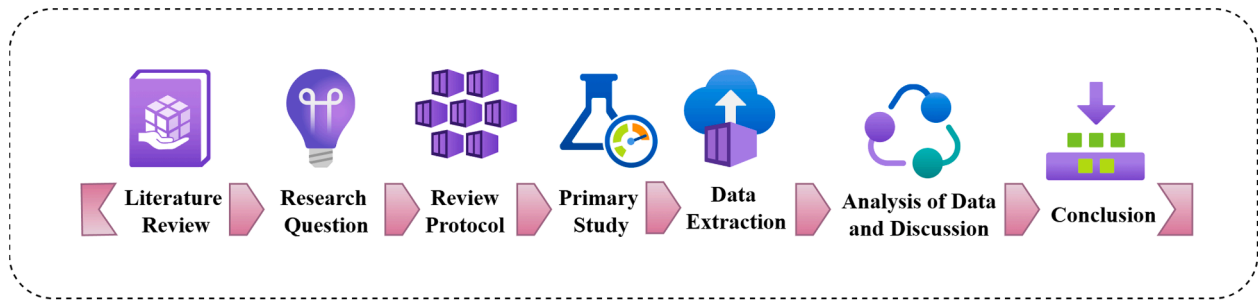


Fig. 1. Methodology diagram.

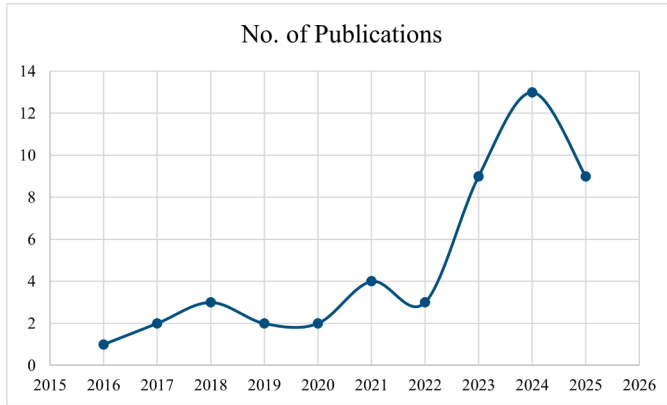


Fig. 2. Number of review publication in past years.

forgery classes; a tick (✓) indicates that the corresponding forgery type or model family is explicitly explained or discussed in the review, while a cross (X) indicates that it is not discussed in the paper.

- [Dehghani and Saberi \(2025\)](#) review deepfake, morphing, and GAN-based image forgeries using traditional, machine-learning, CNN, deep-learning, and transformer detectors.
- [Singh et al. \(2025a\)](#) focus on deepfake and GAN-generated forgeries, comparing machine-learning, CNN, and deep-learning models rather than providing a structured coverage of classical forgery families or datasets
- [Kumari and Garg \(2025\)](#) survey copy-move and splicing forgeries with traditional, machine-learning, CNN, and deep-learning approaches and commonly used datasets.
- [Kaur et al. \(2024\)](#) examine copy-move, splicing, and inpainting forgeries using traditional, machine-learning, CNN, and deep-learning models, with analysis of datasets.
- [Muthaiah et al. \(2024\)](#) provide a review of deepfake, morphing, and GAN-based forgeries, focusing on CNN and transformer architectures and representative datasets.
- [Passos et al. \(2024\)](#) analyze deepfake, morphing, and other GAN-based forgeries using machine-learning, CNN, and transformer detectors together with datasets.
- [Singh and Kumar \(2024\)](#) cover copy-move, splicing, inpainting, morphing, and GAN-based forgeries with traditional, machine-learning, CNN, and deep-learning models.
- [Barglazan et al. \(2024\)](#) survey copy-move, splicing, inpainting, and GAN-based forgeries across all model families (traditional, ML, CNN, DL, transformers) and datasets.
- [Edwards et al. \(2024\)](#) focus on deepfake and GAN-generated forgeries, reviewing machine-learning, CNN, and transformer-based detectors and datasets.
- [Heidari et al. \(2024\)](#) primarily discuss deepfake, morphing, and GAN-based forgeries using machine-learning, CNN, and deep-

learning models, with little emphasis on datasets or classical copy-move / splicing forgeries.

- [Gupta et al. \(2023\)](#) review deepfake and GAN-based image forgeries spanning traditional, machine-learning, CNN, and deep-learning detectors and datasets.
- [Pandey et al. \(2023\)](#) analyze copy-move, splicing, and inpainting forgeries mainly from a deep-learning perspective, reporting datasets.
- [Shi et al. \(2023\)](#) review copy-move, splicing, inpainting, and GAN-based forgeries using traditional, CNN, and deep-learning models and datasets.
- [Kaur et al. \(2023\)](#) provide a method-oriented survey of traditional, machine-learning, and CNN-based forensic detectors and datasets.
- [Castillo Camacho and Wang \(2021\)](#) cover copy-move, splicing, deepfake, and inpainting forgeries and compare traditional, machine-learning, and CNN-based detectors and datasets.
- [Ferreira et al. \(2020\)](#) focus mainly on copy-move forgery detection with traditional and deep-learning approaches.
- [Thakur and Rohilla \(2020\)](#) review copy-move, splicing, morphing, and GAN-based forgeries using traditional and CNN-based detectors and datasets.
- [Rani and Kumar \(2019\)](#) summarize copy-move, splicing, and inpainting detection methods based largely on traditional models.
- [Teerakanok and Uehara \(2019\)](#) survey copy-move forgery detection using traditional, machine-learning, and CNN-based models, focusing on algorithms.
- [Dixit and Gupta \(2016\)](#) provide an early review of copy-move and splicing detection using traditional models, without deep-learning, transformer, GAN-based, or dataset-level analysis.
- *Our Work* offers a unified survey that jointly covers all major forgery families (copy-move, splicing, deepfake, inpainting, morphing / retouching, and GAN-based forgeries), all model paradigms (traditional, ML, CNN, general deep learning, and transformers), and dataset usage and evaluation protocols in a single framework.

2.1. Digital image forgery

Digital image forgery refers to any alteration of an image intended to deceive the viewer or misrepresent the truth. Common forms of forgery include copy-move forgeries, image splicing, image inpainting, and enhancement / retouching. In a copy-move forgery, a region of the image itself is copied and pasted elsewhere in the same image to conceal or duplicate an object ([Farhan et al., 2024](#)). In an image splicing forgery, content from one image is inserted into another image, creating a composite that did not exist in reality ([Dutta et al., 2025](#)). Inpainting of images or content removal involves erasing an object or region and filling it in using computer-generated content ([Barglazan et al., 2024](#)). Retouching includes subtle modifications, such as adjusting color, brightness, or applying filters, which may not create obvious inconsistencies but can still mislead. In recent years, GAN-based forgeries have emerged in which a generative adversarial network (GAN) ([Mehrpour et al., 2023](#)) or other deep-generative model produces synthetic imagery or modifies existing

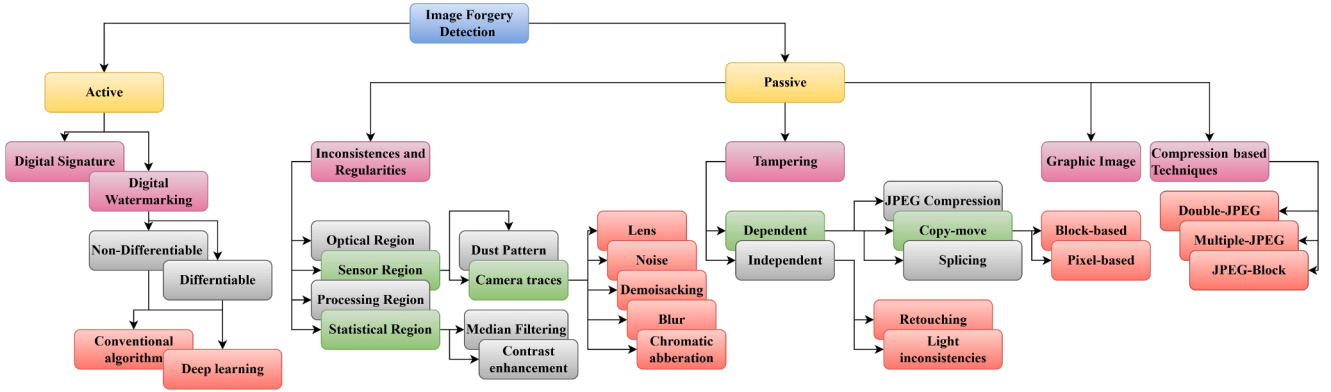


Fig. 3. Hierarchical taxonomy of image-forgery detection and localization.

images. These AI-generated forgeries often span the entire image or a large region and can be very photorealistic (Zanardelli et al., 2023).

2.2. Active vs passive forensics

Techniques for verifying image integrity are often categorized as active or passive. In active forensic techniques, some form of information is embedded in the image at the time of capture or before distribution, for example, a digital watermark or signature, which can later be verified for integrity (Thakur & Rohilla, 2020). If the watermark is missing or altered, the image is deemed inauthentic. Active approaches require specialized hardware or prior embedding and are not applicable to most images found 'in the wild' that lack such protection. Passive (blind) forensic techniques, in contrast, assume no prior embedded data and instead analyze the inherent statistics of the image to detect tampering (Bargazan et al., 2024).

Active approaches verify integrity using preattached evidence, digital signatures, and watermarking, implemented through non-differentiable classical algorithms or differentiable learning-based schemes. Passive approaches require no prior marks and infer manipulation from intrinsic cues grouped into four evidence families: optical / lens effects, sensor traces, processing artifacts (median filtering, blur), and statistical / photometric irregularities (contrast, chromatic, or illumination inconsistencies). The taxonomy also enumerates the principal manipulation classes targeted by detectors, JPEG compression traces, copy-move (with block- and pixel-based pipelines), splicing, and retouching, while isolating compression-specific indicators (double / multiple JPEG, block artifacts) and a branch for non-photographic / graphic imagery. Collectively, Fig. 3 clarifies where detection / localization methods obtain forensic signals and how those signals map to specific tampering categories.

2.3. Forgery detection vs localization

Image forgery detection generally refers to the determination of whether an image has been manipulated (a binary decision: authentic vs. tampered) (Thakur & Rohilla, 2020). Muthaiah et al. (2024) summarize how to distinguish real photos from AI-generated ones, highlighting the scope, key datasets, and evaluation focus. It explains the major generators (GANs, Stable Diffusion, DALL-E) and surveys detectors from CNNs to transformers (residual network (ResNet), visual geometry group (VGG), EfficientNet, DenseNet, shifted window transformer (Swin-T)). Findings: detectors score very high on same-generator data but drop sharply with cross-generator tests or compression; the paper suggests robust designs like EfficientNet and ensembles. Forgery localization goes a step further by identifying which pixels or regions have been altered. Localization is typically represented as a binary mask the same size as the image, where 1 indicates tampered regions and 0 indicates pristine regions. In copy-move forgeries (Dixit & Gupta, 2016),

some methods further distinguish between the source region and the target region. This can be treated as a three-class segmentation: authentic, source, target. Accurate forgery segmentation is particularly important in forensic analysis, as it can highlight exactly what part of an image has been manipulated, facilitating human interpretation and evidence use (Ansari et al., 2014).

3. Screening methodology and taxonomy

We frame this review using a review-like structure, planning, implementation, and reporting, focused here on the planning stage for image forgery detection and localization (IFDL) Fig. 1. The scope targets still-image manipulations spanning copy-move, splicing, removal / inpainting, and fully synthesized content, and organizes evidence into four families: RGB / statistical residuals, compression-resampling artifacts, sensor / camera fingerprints, and physics / geometry inconsistencies, alongside modern learning-based and hybrid pipelines that fuse these cues for pixel-level masks. We synthesize primary work from 2016 to 2025, emphasizing advances that improve boundary fidelity, source-target disambiguation, and robustness to OSN-style degradations (JPEG, resizing, blur), while noting persistent gaps on tiny / textureless edits and out-of-distribution generalization. Practical guidance is provided on datasets (e.g., CASIA, NIST16, COVERAGE, Columbia, IMD; face / deepfake suites), metrics (area under the curve (AUC) / F1 / accuracy (ACC), interaction over union (IoU)), and evaluation protocols (cross-dataset and online social network (OSN)-degraded testing). Together, the review establishes a contemporary foundation for our image forgery detection and location framework and its implementation in the real-world quality assurance of digital media.

3.1. Search strategy

We conducted a comprehensive literature search that spanned 2016 to 2025 to gather relevant publications on image forgery detection and localization. Our search covered academic databases using keywords such as image forgery detection, image tampering localization, copy-move forgery, splicing detection, deepfake detection, and image forensics. After defining the initial keyword list, we manually reviewed the terms and expanded them with common synonyms, alternative spellings, and broader descriptors to avoid missing relevant work. For each database, the final search strings combined a problem block (e.g., "image forgery detection", "image tampering localization", "copy-move forgery", "deepfake detection") with a method block (e.g., "traditional method" "deep learning", "convolutional neural network", "vision transformer") using Boolean operators. Within a block, related terms were linked by OR, while the blocks themselves were joined by AND to restrict results to genuinely relevant studies. Whenever supported (for example, in IEEE Xplore), wildcard operators such as "forg*" were also

Table 3
Quality appraisal criteria for primary studies (IFDL).

Type	Definition
Internal	The study clearly states its goals, scope, assumptions, threat model, and experimental setup for IFDL (manipulation types, datasets, baselines, and preprocessing).
External	Findings demonstrate generalizability beyond the training domain (e.g., cross-dataset / OSN tests, unseen tools/models, different resolutions / compressions).
Construct	Research questions are mapped to appropriate measures and design: image- vs. pixel-level tasks, correct metrics (AUC, F1, ACC, IoU), sound ablations and statistics.
Conclusion	Claims are supported by evidence with limitations / error analysis, and provide reproducibility signals (code / data / weights or sufficient implementation detail).

Table 4
Data extraction schema used in this review.

Purpose	Meta-data
General information	Title; authors; venue; year; digital object identifier (DOI) / uniform resource locator (URL).
Specific details	Manipulation scope (CM, splicing, removal, synthetic); cues used (RGB / statistical, resampling / compression, photo-response non-uniformity (PRNU) / color filter array (CFA) / conditional random field (CRF), physics / geometry); model / architecture (CNN / Transformer / hybrid), supervision and losses; datasets and splits; metrics (AUC, F1, ACC, IoU, boundary scores); robustness tests (JPEG, resize, blur, OSN); cross-dataset results; runtime frames per second (FPS), memory, input size; code / weights availability; failure modes / limitations.

used, so that a single query matched variants like "forgery", "forged", and "forgeries" in line with the database's search syntax. Primary searches targeted scholarly sources (IEEE Xplore, ACM DL, SpringerLink, ScienceDirect, Elsevier, Wiley, MDPI, SPIE, arXiv) throughout 2016–2025; conference proceedings, journals and relevant literature were included. Google and Google Scholar helped find leads, but were not used to count unique studies. Secondary screening (titles, abstracts, conclusions) and snowballing of reference lists completed the coverage. 588 articles were identified and selected. We prioritized peer-reviewed journal and conference papers, while also including a few influential preprints that introduced novel ideas. To ensure relevance, we include only works that focus on digital image forgeries. Each paper was evaluated for technical soundness and contribution. We excluded papers that only dealt with watermarking (active methods) or steganography, since our focus is on passive detection of visual tampering. After initial selection, approximately 120 articles were selected as core references for this review. Then these papers were grouped by approach and forgery type, forming the basis of our taxonomy.

- **Inclusion:** English, peer-reviewed (or archival preprints with clear methods), direct relevance to image forgery detection and localization, and empirical results from academic or applied settings.
- **Exclusion:** Non-English, unavailable full text, purely theoretical pieces without experiments, or works outside of image forensics.
- **Quality assessment:** Each candidate was evaluated on four validity axes (see Table 3): internal (clarity of aims / assumptions for IFDL), external (applicability beyond the lab, including cross-dataset or OSN tests), construct (sound mapping between research questions, measures, and outcomes) and conclusion (evidence-based claims consistent with results). Snowball checks on accepted papers further tested relevance and completeness.
- **Data extraction:** We used structured Excel forms to capture metadata and technical factors (see Table 4): venue / year, manipulation scope, cues used (RGB / noise / frequency / physics), model type and supervision, datasets (e.g., CASIA, NIST16, Coverage, Columbia, IMD, face / deepfake suites), metrics (AUC, F1, Acc, IoU, boundary quality),

Table 5
Empirical study types and brief explanations (IFDL).

Type of Study	Explanation
Case Investigation	Goal-driven evaluation on a concrete scenario or dataset (e.g., newsroom / platform pipeline, specific manipulation family).
Industry Assessment	Practice-based account from operational settings without predefined hypotheses; emphasizes constraints and deployment findings.
Experimental Study	Controlled experiments with stated protocols, baselines, and ablations to compare IFDL methods under matched conditions.
Action Research	Method co-designed and iteratively refined within a live workflow; joint development and evaluation with stakeholders.
Survey	Structured evidence gathering (questionnaires, interviews, benchmark meta-analysis) summarizing practices, datasets, or results.
Not Defined	Methodology is insufficiently described to assign a category.

training size, code / weights availability, runtime, robustness tests (JPEG / resizing / blur), and generalization (cross-dataset / OSN).

- **Empirical study:** Studies were labeled per Table 5: case investigations (objective-driven evaluations in specific scenarios), industry evaluation (practice-driven reports without predefined hypotheses), experimental studies (controlled comparisons under stated protocols), action research (method development embedded in a real pipeline), surveys (quantitative / qualitative evidence gathering) and not defined (insufficient methodological detail).

Methodologically, this work follows PRISMA-style reporting but should be interpreted as a structured, PRISMA-guided literature review rather than a fully pre-registered systematic review. Study selection was carried out by two reviewers (the first author and the supervising professor (co-author)). After de-duplication, both reviewers screened titles and abstracts using the inclusion and exclusion criteria in Table 3 and Table 4, and potentially eligible papers were then assessed in full text against the quality criteria in Table 3 and the empirical study types in Table 5. Any disagreements at either stage were resolved through discussion until consensus was reached. The validity dimensions and study-type labels were used qualitatively to support inclusion and exclusion decisions and to interpret the strength of the evidence base, rather than to produce separate numerical counts of how many papers satisfied each individual criterion.

In line with best practices for systematic reviews, our screening process is summarized in three complementary figures. Fig. 4 (PRISMA-style flow) traces the selection pipeline from initial retrieval to final inclusion, while N represents the numbers of total and remaining articles: 588 records were identified, 396 remained after duplicate removal, 270 after miscellaneous filters (non-English, out-of-scope titles, and inaccessible items), 148 after abstract screening, and 120 articles were ultimately retained following full-text review. To contextualize the evidence base, Fig. 5 (venue distribution) shows that the included studies are predominantly journal articles (80%), with the remainder conference papers (20%), reflecting the field's maturity and emphasis on archival publication for detailed methodological reporting. Table 6 and Fig. 6 (publisher breakdown) indicates that the literature is concentrated among a few major publishers, IEEE (30%), Elsevier (21%), Springer (16%), MDPI (12%), Wiley (10%), with smaller contributions from ACM (5%), Taylor & Francis (3%), SSRN (2%), and SAI (1%), providing a transparent view of source diversity and potential publication-venue biases. Finally, Fig. 7 represents the distribution of the publication in the past years.

3.2. Taxonomy of forgery techniques

We categorize existing methods along two primary dimensions: (1) the technical approach and (2) the type of forgery addressed.

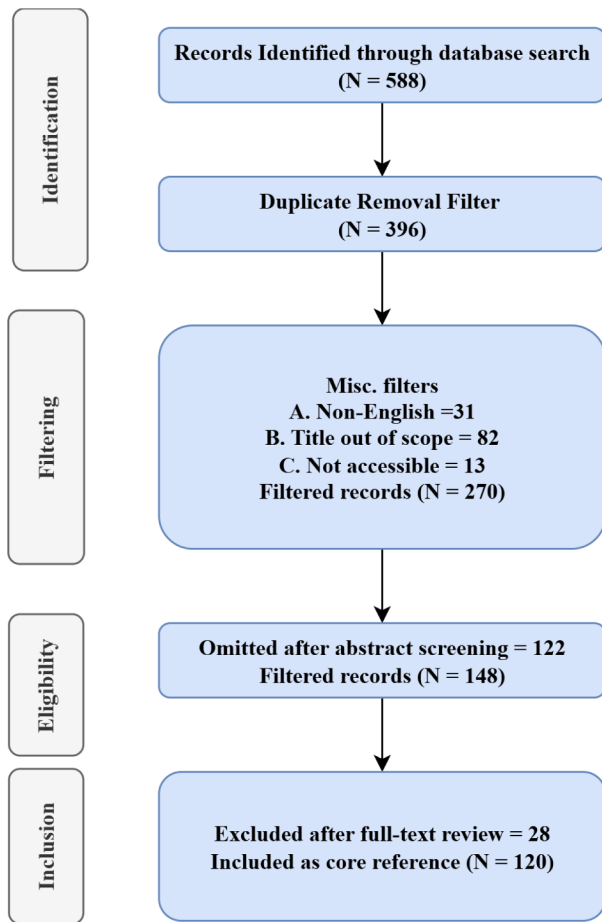


Fig. 4. PRISMA-style screening flow diagram.

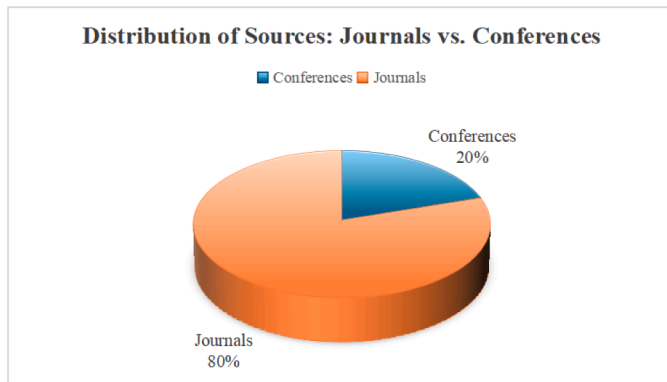


Fig. 5. Distribution of included studies by publication venue, Journals (80%) vs. Conferences (20%), 2016-2025.

• Technical Approach:

- **Traditional Techniques:** Methods that rely on manually engineered features and classical machine learning. These include signal processing algorithms, statistical analyses, and shallow learning models. [Section 5.1](#) covers this category.
- **CNN-Based Techniques:** Methods that use convolutional neural networks to automatically learn discriminative features for forgery detection ([Section 5.2](#)). These range from classification CNNs that detect if an image (or patch) is manipulated to fully convolutional networks that localize tampered regions at pixel level.

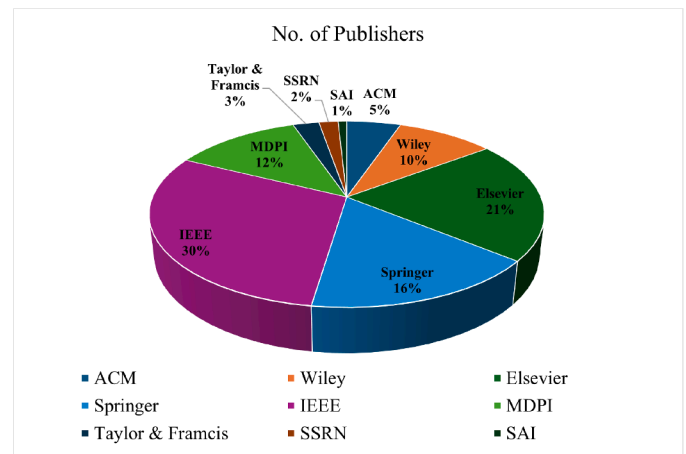


Fig. 6. Publisher distribution of articles, IEEE (30%), Elsevier (21%), Springer (16%), MDPI (12%), Wiley (10%), ACM (5%), Taylor & Francis (3%), SSRN (2%), and SAI (1%) for 2016-2025.

- **Transformer-Based Techniques:** Recent methods that incorporate vision transformers or self-attention mechanisms to capture long-range dependencies in images ([Section 5.3](#)). These aim to improve on CNNs by modeling global context, or are used in hybrid CNN-Transformer architectures.
- **GAN and Multimodal Techniques:** Approaches that use generative models to aid detection or combine multiple input modalities ([Section 5.2](#)). GAN-based detectors may involve adversarial training or one-class models, whereas multimodal methods fuse information beyond the image pixels (e.g., video frame + audio analysis, or image + metadata).

• Forgery Type:

- **Copy-Move Forgery:** A part of the image is copied and pasted onto another area of the same image. The goal is usually to conceal an object or duplicate a pattern (e.g., cloning out a person by covering them with a copied background patch). Copy-move is one of the most prevalent tampering types. Detection often involves finding duplicate regions through feature matching.
- **Image Splicing:** Combining two or more *different* images into a seamless composite. For example, inserting a person from one photo into another photo. Splicing is considered more 'aggressive' than retouching and can be very misleading. Detectors look for inconsistencies along splicing boundaries or mismatched camera traces between parts of the image.
- **Object Removal (Image Inpainting):** This is a special case of splicing where a region is removed and replaced by computer-generated filler to blend with the background. Modern techniques *inpainting* (often GAN-based) can produce very realistic results, making detection challenging.
- **Image Retouching:** Subtle enhancement or suppression of features (e.g., smoothing of the skin, adjusting brightness). Retouching does not add or remove content but changes the visual appearance. It is generally considered a mild form of forgery, but excessive retouching (e.g., in a propaganda image) can still mislead viewers. Detecting retouching is difficult as changes are global and low-magnitude (often requiring detection of statistical irregularities in histograms, noise, or compression artifacts).
- **AI-Generated Forgeries (Deepfakes):** Although technically a form of splicing or synthesis, the rise of *deepfakes* (particularly face swaps and entirely AI-generated images) warrants separate mention. These involve neural networks (GANs or Variational autoencoders (VAEs)) creating or altering faces and objects in images. They often require specialized detection approaches (for example, focusing on physiological signals in faces or frequency

Table 6
Summary of journals, conferences, and database used to retrieve research articles.

Journals, Conferences, and Database Details	Publisher	Total Count
Journals Details		
Expert Systems	Wiley	29
WIREs Data Mining and Knowledge Discovery	Wiley	27
Journal of King Saud University - Computer and Information Sciences	Elsevier	4
Journal of Visual Communication and Image Representation	Elsevier	6
Signal Processing	Elsevier	18
Pattern Recognition	Elsevier	17
Computers and Electrical Engineering	Elsevier	13
Forensic Science International	Elsevier	12
Expert Systems with Applications	Elsevier	4
Data and Information Management	Elsevier	7
Computers in Human Behavior Reports	Elsevier	7
Signal Processing: Image Communication	Elsevier	10
Neurocomputing	Elsevier	8
Multimedia Tools and Applications	Springer	8
International Journal of Computer Vision	Springer	5
Multimedia Systems	Springer	17
MultiMedia Modeling	Springer	14
Artificial Intelligence Review	Springer	8
Wireless Personal Communications	Springer	14
Journal of Computational Social Science	Springer	8
IEEE Access	IEEE	12
IEEE Transactions on Information Forensics and Security	IEEE	10
IEEE Transactions on Image Processing	IEEE	16
IEEE Transactions on Multimedia	IEEE	18
IEEE Transactions on Circuits and Systems for Video Technology	IEEE	15
IEEE Transactions on Pattern Analysis and Machine Intelligence	IEEE	17
Remote Sensing	MDPI	14
Electronics	MDPI	12
Journal of Imaging	MDPI	18
Mathematics	MDPI	10
Applied Sciences	MDPI	19
ACM Transactions on Multimedia Computing, Communications, and Applications	ACM	31
Journal of Cyber Security Technology	Taylor & Francis	15
International Journal of Advances in Intelligent Informatics	Universitas Ahmad Dahlan	13
Conferences Details		
Computer Vision - ECCV 2018	Springer	10
Proceedings of the International Conference on Innovative Computing & Communication (ICICC)	SSRN	11
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)	IEEE	24
International Conference on Image Processing and Media Computing	IEEE	8
IEEE/CVF International Conference on Computer Vision (ICCV)	IEEE	9
IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)	IEEE	11
IEEE Winter Conference on Applications of Computer Vision (WACV)	IEEE	8
International Conference on Ubiquitous Computing and Intelligent Information Systems (ICUIS)	IEEE	5
Proceedings of the Third International Conference on Electronics Communication and Aerospace Technology (ICECA)	IEEE	6
International Conference on Computing, Communication, and Intelligent Systems (ICCCIS)	IEEE	8
International Conference on Image Information Processing (ICIIP)	IEEE	4
International Conference on Information and Automation	IEEE	5
International Journal of Advanced Computer Science and Applications (IJACSA)	SAI	5
Database Details		
ScienceDirect	Elsevier	18

artifacts of generative models). Deepfakes fall under passive detection, but represent a new frontier due to the sophistication of AI generation.

4. Dataset details

Digital image forgery detection and localization research is heavily based on diverse and representative datasets for training and evaluation. In the past two decades, many datasets have been created to facilitate the development of algorithms that can detect whether an image has been manipulated and localize the tampered regions. The types of forgeries of interest typically include image splicing (cut-and-paste of content between images), copy-move forgeries (copying and pasting content within the same image), image retouching or enhancement (adjustments of colors, filters, etc.), and object removal (inpainting) where a region is filled to hide content. Early 'classical' datasets were relatively small and often manually created, suitable for evaluating specific algorithms. With the rise of deep learning, much larger datasets (sometimes

automatically generated) have emerged to train data-hungry models. There are also hybrid datasets that combine real-world manipulations with synthetic ones.

we compile a comprehensive list of the major datasets used for image forgery detection and localization. We organize the datasets by the primary forgery type or use case they address.

4.1. Datasets for copy-move forgery detection

Copy-move forgery (cloning part of an image into another area of the same image) is a common manipulation, and several datasets have been specifically created for developing and evaluating copy-move detection algorithms:

- *MICC F220, F600, F2000 (2011)*: Early copy-move benchmarks: 220 / 600 / 2000 images (tampered / original counts: 110 / 110, 160 / 440, 700 / 1300). JPEG, up to HD. Tampering via intra-image copy with rotation / scaling. No pixel-level masks (mainly binary

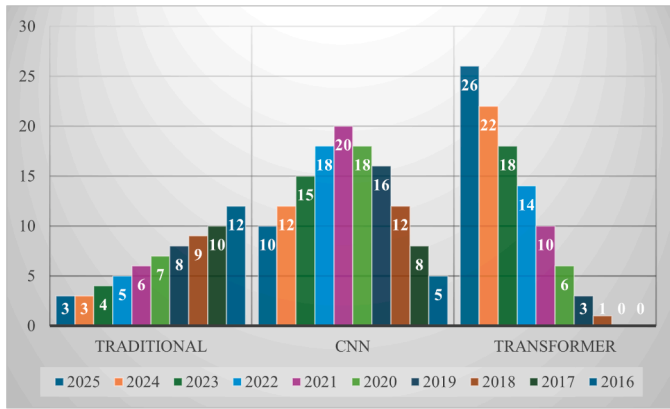


Fig. 7. Yearly distribution of publications.

detection). Released with a SIFT-based CMFD method (Amerini et al., 2011).

- **CoMoFoD (2013)**: Comprehensive CMFD set: 260 forged sets; each set has original + tampered + masks (separate source and target). Two resolutions: $200 \times (512 \times 512)$ and $60 \times (\text{up to } 3000 \times 2000)$. Includes scaling, rotation, noise, blur (Tralic et al., 2013).
- **COVERAGE (2016)**: Self-similarity-challenging CMFD: 100 authentic + 100 tampered pairs; medium res (400×486). Each tampered image ships with its original and a binary duplication mask (Wen et al., 2016).
- **FAU Copy-Move (2012)**: Christlein et al. (2012) evaluated several popular copy-move detection approaches on a dataset of 48 base images with multiple tampered versions. This small set (sometimes referenced as the FAU dataset, from University of Erlangen-Nuremberg) contains 48 tampered images and was primarily used to compare algorithm robustness. Although not a widely distributed standalone dataset, it contributed to early benchmark comparisons.
- **GRIP and Others**: A few other copy-move datasets, such as the GRIP dataset (Cuzzolino et al., 2015), but these are less commonly used. GRIP, for example, was introduced with a dense-field detection method. In practice, CoMoFoD, MICC, and COVERAGE remain the go-to benchmarks for copy-move forgery detection research.

4.2. Datasets for image splicing detection

Image splicing (also known as photomontage or cut-and-paste forgery) involves inserting content from one image into another. This often introduces inconsistencies (lighting, noise, edges) that splicing detection algorithms try to catch. Key datasets include:

- **Columbia Uncompressed Image Splicing Dataset (2004)**: One of the earliest splicing datasets, introduced by Hsu and Chang (Columbia University DVMM Lab). It contains 180 spliced images and 183 authentic images, stored in uncompressed TIFF / BMP format. This dataset was distributed as pairs of authentic and tampered images, and splicing locations were typically known to the creators, although no explicit mask files were initially released. The Columbia dataset was a staple in early papers; for example, it was used to evaluate camera consistency and geometry-based splicing detectors (Ng et al., 2004). Researchers sometimes refer to a smaller dataset of 'Columbia image blocks' (1845 image patches of size 128×128), but the uncompressed splicing dataset with full images (also called the Columbia Color dataset) is the primary one used for forgery detection. It is available on request from the Columbia website (Hsu & Chang, 2006).
- **CASIA Image Tampering Detection Evaluation Database v1.0 (2009) and v2.0 (2013)**: The CASIA datasets, released by the Chinese Academy

of Sciences Institute of Automation (CASIA), have been extremely influential. CASIA v1.0 contains 921 tampered color images and 800 authentic images. All forgeries in v1.0 are splicing-type (objects or regions from other images added or removed) and the images are about 384×256 pixels in JPEG / TIFF format. CASIA v2.0 is much larger, with 5123 tampered images and 7491 authentic images. CASIA v2 includes a mix of splicing and copy-move forgeries, as well as some image enhancement / retouching (Dong et al., 2013).

- **DSO-1 (Carvalho's dataset, 2013)**: DSO-1 dataset was introduced by De Carvalho et al. (2013) for illumination-based splicing detection. It consists of 200 images: 100 authentic and 100 spliced photographs. The images are high-resolution (mostly 2048×1536) and realistic, each spliced image was skillfully created by inserting a person or object from one image into another with consistent quality. The splices are meant to be visually plausible (consistent lighting, etc. to challenge detection). All images are in JPEG format. Ground truth masks indicating the spliced region were created (since the splices were manual, the authors knew the exact pasted region), though the dataset release primarily provides the images labeled authentic or tampered. This dataset is often called the Carvalho's dataset or DSO-1.
- **Realistic Tampering Dataset (Korus, 2016)**: This dataset by Korus and Huang (2016a) contains 220 high-resolution forged images along with their originals and multi-level ground truth masks. Each tampered image (1920×1080 TIFF) was created manually using photo editing software (GIMP / Affinity) to simulate realistic manipulations involving object insertion or object removal.
- **Wild Web Tampered Images (Zampoglou, 2015)**: Zampoglou et al. (2015) compiled a collection of spliced images 'in the wild' from the Internet. Often referred to as the Web Segmentation dataset or Wild Web, it contains 9657 real-world tampered images gathered from online sources (e.g. fake images circulating on the web) and a small set of 90 true (untampered) images for reference.

4.3. Datasets with multiple manipulation types

As research progressed, datasets were created to include various types of forgeries in one collection, reflecting practical scenarios where an image could undergo any kind of manipulation. These comprehensive datasets serve as benchmarks for general forgery detection and localization, often used in challenges and competitions.

- **NIST Nimble Challenge 2016 (NIST16)**: The Nimble Challenge series, organized by NIST, produced evaluation datasets for image forensic algorithms. NIST16 is a dataset of 564 high-resolution manipulated images covering three tampering categories: splicing, copy-move, and object removal (inpainting) (Guan et al., 2019).
- **Open Media Forensics Challenge Datasets (2018-2022)**: Following the Nimble challenges, the Media Forensics Challenge (MFC) program continued to produce large datasets. For example, the OpenMFC 2020 Image Manipulation dataset contains more than 16,000 manipulated images spanning splices, copy-moves, and removals (Guan et al., 2019).
- **Defacto Dataset (2019)**: The DEFACITO dataset was introduced by Mahfoudi et al. (2019) as a large benchmark for multiple forensic tasks. It contains on the order of 190,000 manipulated images covering splicing, copy-move, and removal forgeries, created automatically by applying these manipulations to images from the MS-COCO dataset.
- **Fantastic Reality (NeurIPS 2019)**: Fantastic Reality is a dataset released by an NVIDIA research team (Kniaz et al., 2019). In their paper 'The Point where Reality Meets Fantasy'. It includes 16,000 authentic and 19,000 tampered images of various scenes, with only splicing forgeries. Fantastic Reality is often used to evaluate the deep learning model's ability to generalize, as it provides large-scale, high-quality splices with ground truth.

Table 7

Summary of datasets used in forgery detection and localization.

Dataset (Year)	Forgery Type(s)	Size (Tampered / Authentic)	Ground Truth Annotation	Citation (by Sep 2025)	Download Link
tampCOCO (Kwon et al., 2021a)	Splicing, Copy-Move (auto)	800,000 tampered	Yes (mask per tamper)	218	link
DF2023 (Fischinger & Boyer, 2023)	Splice, Copy-Move, Removal, Enhancement	1,000,000 tampered (train) + 5k val	Yes (pixel masks)	1	link
MISD (Kadam et al., 2021)	Multiple splices per image	300 tampered (multi-object)	Yes (mask for each splice)	28	link
IMD2020 (Novozamsky et al., 2020)	Splice, Copy-Move, Removal	35k real; 35k inpainted; 2k real fakes	Yes (masks for all manipulations)	265	link
DFDC (Dolhansky et al., 2020)	DeepFake	1131 / 4113 (videos)	No	1198	link
DEFACTO (Mahfoudi et al., 2019)	Splice, Copy-Move, Removal + Faces	~189k tampered (149k general, 40k faces)	Yes (pixel masks for all)	161	link
Fantastic Reality (Kniaz et al., 2019)	Splicing (GAN-blended)	19,423 / 16,592	Yes (mask, instance labels)	95	link
FaceForensics++ (Rössler et al., 2019)	DeepFake	1000 / 5000 (videos)	No	3365	link
UADF (Li et al., 2018)	DeepFake	49 / 49 (videos)	No	1094	link
PS-Battles (Heller et al., 2018)	Mixed (splice, copy-move)	~90k / 11k	No	63	link
COVERAGE (Wen et al., 2016)	Copy-Move	100 / 100	Yes (binary mask)	483	link
NIST16 (Guan et al., 2019)	Splice, Copy-Move, Removal	564 tampered (high-res)	Yes (pixel masks)	332	NC17; MFC18
Realistic Tampering (Korus & Huang, 2016a)	Object insertion & removal	220 / 220	Yes (multi-class mask)	242	link
In the Wild (Huh et al., 2018)	Splicing	201 tampered	Yes	578	link
CASIA v2.0 (Dong et al., 2013)	Splicing, Copy-Move, Retouch	5,123 / 7,491	No (unofficial masks exist)	897	link
DSO-1 Carvalho (De Carvalho et al., 2013)	Splicing (realistic)	100 / 100	Yes (known splice region)	467	link
CoMoFoD (Tralic et al., 2013)	Copy-Move	260 tampered sets	Yes (source & target masks)	494	link
MICC F220, F600, F2000 (Amerini et al., 2011)	Copy-Move	110 / 160 / 700 tampered (in 3 sets)	No (region known by design)	1322	link
CASIA v1.0 (Dong et al., 2013)	Splicing (some retouch)	921 / 800	No (labels only)	897	link
Columbia Uncompressed (Hsu & Chang, 2006)	Splicing (photomontage)	180 / 183	No pixel masks (pairs of images)	240	grey; color

- **PS-Battles (2018):** The PS-Battles dataset was collected by Heller et al. (2018) from popular photoshopbattles. This community provides a rich source of 'in the wild' manipulations: for a given original image posted, multiple users contribute their manipulated 'derivative' versions. The PS-Battles dataset comprises 11,142 groups of images, each group having one original and several manipulated versions, totaling 102,028 images.

As summarized in Table 7, the benchmarks used in image-forgery detection / localization span two decades and multiple manipulation families. The catalog ranges from early, small, realistic sets, Columbia Uncompressed and CASIA v1, and classic copy-move / splicing suites (MICC F220 / F600 / F2000, CoMoFoD, COVERAGE, DSO-1) to mixed-manipulation collections with pixel masks (Realistic Tampering, NIST16) and today's large-scale resources built for deep models (PS-Battles, DEFACTO, IMD2020, tampCOCO, DF2023). For face-centric synthesis, video benchmarks such as FaceForensics++ and DFDC are included, typically without pixel-level masks. The table explicitly indicates the level of supervision, none, binary / instance masks, or multi-class masks, since this determines whether methods can be trained and evaluated for localization in addition to detection. Complementing this overview, Fig. 8 illustrates representative triplets, pristine image, tampered counterpart, and ground-truth mask, for three manipulation families (object removal / inpainting, splicing, and copy-move / duplication), clarifying the visual nature of annotations and the segmentation targets used by modern methods.

5. Technical terminologies

5.1. Traditional image forgery techniques

Traditional forgery detection methods based on hand-crafted features and classical machine learning or statistical techniques. These methods dominated the field prior to the deep learning era, and many remain relevant as lightweight or interpretable solutions. Traditional

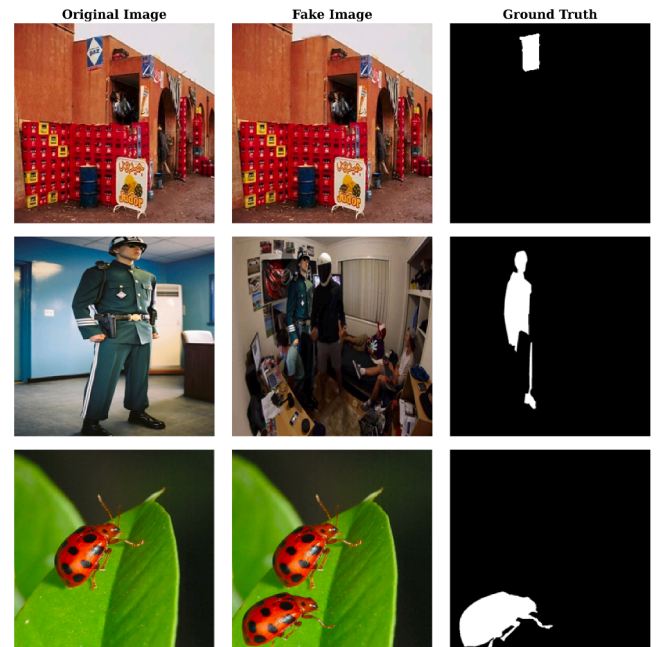


Fig. 8. Illustrative tampered images with their ground-truth masks. Rows show: top; object removal / inpainting, middle; splicing, bottom; copy-move / duplication. Column one contains the pristine image, column two the tampered (fake) counterpart, and column three the binary ground-truth mask marking the altered regions. These examples span diverse manipulation patterns and motivate a general-purpose detection framework.

approaches are typically specialized in detecting specific types of tampering by exploiting particular forensic artifacts left by manipulation. Common steps in these methods include: (1) preprocessing the image

(e.g., converting color spaces or extracting residual signals), (2) computing discriminative features (statistical descriptors, frequency-domain coefficients, etc.), and (3) applying a decision mechanism (such as block matching, clustering, or a classifier like support vector machine (SVM)) to identify inconsistencies. We review traditional methods in two main groups: copy-move forgery detection (CMFD) and splicing detection.

5.1.1. Copy-move forgery detection

[Dixit and Bag \(2021\)](#) propose a fast keypoint-based CMFD: CenSurE detects salient points, the Local Image Permutation Interval Descriptor (LIPID) describes them, matches are found via a k -d tree with Best-Bin-First, then fuzzy c -means clustering, and random sample consensus (RANSAC) refines results. It is robust to rotation, reflection, non-affine, and compound edits (blur, noise, multiple cloning), detects reflected / distorted duplicates, and delivers near-perfect true positives with very few false alarms while beating peers on runtime. The main limitation is the reliance on interest points; smooth or low-texture cloned regions may be missed, yet its speed and broad robustness remain stand-out strengths. [Alhaidery and Taherinia \(2022\)](#) develop a hybrid CMFD that merges keypoint and block cues: speeded-up robust features (SURF) + histogram of oriented gradients (HOG) seed matches, simple linear iterative clustering (SLIC) superpixels mark active areas, and overlapping Zernike-moment blocks within them feed a shallow classifier. By focusing blocks only where needed, it catches smooth-region forgeries missed by keypoint-only methods, stays robust to geometric and compound photometric attacks, and delivers precise localization. Reported results are strong (TPR $\geq 93.75\%$, $\approx 7.25\%$ FP), though accuracy depends on superpixel sizing / segmentation, and the multi-stage pipeline adds complexity. [Lu et al. \(2022\)](#) represent a CMFD fusion pipeline that combines scale-invariant feature transform (SIFT) with log-polar SURF to boost keypoint coverage, matches with generalized 2-nearest-neighbor (g2NN) and RANSAC, and then uses an evolving circular domain clustering (ECDC) region-growing step that expands matched points into circular domains for precise localization. This dual-feature + ECDC design lifts the recall on small or low-contrast forgeries, focuses computation near matches for efficiency, remains robust to rotation / scaling and signal attacks, and reports $\approx 3\%$ higher accuracy than several contemporaries. The trade-off is higher complexity from two extractors and iterative growing, but in return you get more complete, reliable tamper coverage, even for tiny fragments. [Lu et al. \(2022\)](#) propose a hybrid copy-move forgery detector segments the image into superpixel blocks (smooth vs. textured), finds SURF keypoints, encodes them with polar complex exponential transform (PCET), matches features, prunes outliers with RANSAC + filtering, and refines masks via morphology / iteration, combining block- and keypoint-based cues to localize tampering in bright, smooth regions or among similar genuine areas, and resisting rotation, scaling, blur, JPEG compression, and noise. Despite low computational cost and fewer false matches, performance depends on sensitive morphology / segmentation parameters, needs faster matching, and can struggle after extreme down / up-sampling.

[Li and Zhou \(2018\)](#) develop a fast copy-move forgery detector that boosts SIFT keypoints in smooth / small areas by lowering the contrast threshold and rescaling, then performs hierarchical matching with an iterative localization stage that takes advantage of dominant orientation, scale, and color to suppress false alarms; they report stronger F1 / true positive rate (TPR)-false positive rate (FPR) trade-offs than dense field and prior keypoint methods under scaling, rotation, noise, and JPEG compression. [Alberly et al. \(2018\)](#) propose a pragmatic SIFT + Fuzzy C-Means (FCM) pipeline that clusters tentative matches to cut computation; on MICC-220 and a 353 image add-on set, they achieve $\sim 15\%$ lower runtime compared to a traditional SIFT baseline with comparable detection quality, highlighting speed gains without major accuracy loss. [Mohammed et al. \(2018\)](#) Cascading copy-move analysis with a deep resampling detector, CMFD first flags clones; images deemed pristine then pass to a CNN-based resampling framework. In datasets including NIST NC2017 ($\sim 10k$ images), the cascade yields a consistent ~ 8 - 10% gain

in detection rates. [Prasad and Ramkumar \(2016\)](#) compare CMFD using SIFT, SURF, and HOG, plus hybrids (SIFT-HOG and SURF-HOG), finding that hybrids, especially SIFT-HOG, deliver the best overall accuracy / recall, while SURF alone is fastest.

5.1.2. Splicing and general tampering detection

[Korus and Huang \(2016b\)](#) introduce multi-scale PRNU-based tampering localization that fuses correlation maps from multiple window sizes and augments them with content-aware neighborhood interactions to better propagate decisions. They also explore segmentation-guided and adaptive window variants with prior CRF / markov random field (MRF), consistently outperforming single-scale PRNU in small forgeries. [Zampoglou et al. \(2017\)](#) run a large benchmark of splicing-localization methods on web-style imagery, probing robustness to JPEG re-compression and rescaling; their study surfaces systematic weaknesses and releases an open framework for reproducible comparisons. [Liu and Pun \(2020\)](#) estimate local noise via an adaptive singular value decomposition (SVD) scheme and design a Vicinity Noise Descriptor, classifying regions with an SVM; experiments on multi-object splices show higher precision than prior noise-difference approaches, which is crucial for reducing false positives. [He et al. \(2012\)](#) propose a blind splicing detector that builds a large feature set from expanded Markov features in the discrete cosine transform (DCT) domain and new Markov features in the discrete wavelet transform (DWT) domain that model dependencies across positions, scales, and orientations; the features are pruned with SVM-RFE and classified with an SVM. [Chen et al. \(2017a\)](#) detect forged (software) blur along edges by exploiting the non-commutativity of optical blur and camera response functions (CRFs), converting the edge intensity-gradient relationship into a CNN recognition task, and localizing spliced / retouched boundaries more reliably than prior blur consistency cues. [Chen et al. \(2022b\)](#) propose a training-free generalized likelihood ratio test (GLRT) based on a simplified heteroscedastic JPEG noise model whose two parameters act as camera 'fingerprints'; the detector admits analytic control of false alarm rates and achieves competitive splicing detection on TIFF / JPEG without large training sets. [Bondi et al. \(2017\)](#) extract CNN features from the camera model patchwise and cluster them to mark regions originating from different devices; on 2000 images, the approach improves tamper detection / localization against several content-agnostic baselines. [Sekhar and Shankar \(2020\)](#) leverage JPEG block artifact grids but operate at the object level, using cosine dissimilarity across segmented objects to pinpoint the manipulated one in double-compressed images; their object-centric block artifact grid (BAG) variant reports superior localization versus whole-image DCT heuristics.

[Table 8](#) synthesizes mainstream copy-move and splicing detectors across four broad lines: (i) keypoint / hybrid descriptors such as SIFT / SURF + HOG, hierarchical feature-point matching, and superpixel-aided SURF / PCET; (ii) clustering and fusion pipelines (e.g., SIFT + FCM and CMFD combined with resampling cues); (iii) learning-based models that exploit ConvLSTM / CNNs, CRF / blur-edge consistency, and camera-model CNN clustering; and (iv) noise / physics priors including GLRT for heteroscedastic JPEG noise, adaptive-SVD local-noise features, object-wise BAG, and multi-scale PRNU fusion. The methods span image-level decisions and pixel-level masks, with strengths that include robustness to scaling / rotation, illumination changes, and JPEG re-compression, plus speed gains via clustering, domain-evolving clustering, and object-level grouping. Recurring limitations are multi-stage complexity and heavy parameter tuning; reliance on hand-crafted features or RANSAC that struggle on smooth / repetitive textures; binary outputs without dense masks; and sensitivity to double-JPEG, resampling, or color shifts. A clear trend favors hybridization, combining complementary cues and multiscale reasoning, to secure higher TPR and tighter localization at acceptable runtime. Learning-based approaches improve boundary localization but depend on data and may weaken across devices, while training-free statistics offer controllable false alarms, yet can miss small or flat-region forgeries.

Table 8
Summary of traditional copy-move and splicing detection methods.

Ref	Method	Features used	Detection Type	Strengths	Limitations
Alhaidery and Taherinia (2022)	SURF-HOG + Zernike	SURF + HOG (stage 1); SLIC superpixel segmentation; Zernike moments (stage 2)	Copy-Move (hybrid)	Adapts to texture; refines smooth regions; TPR $\geq 93.8\%$, FPR $\sim 7\%$; robust to geometric & photometric attacks	Multi-stage complexity; depends on good segmentation; mis-set superpixels may under-detect
Lu et al. (2022)	SIFT + log-SURF + ECDC	SIFT + log-polar SURF; g2NN; evolving circular domain clustering (ECDC)	Copy-Move (fusion)	Fuses features (smooth & textured); ECDC fills gaps; robust to transforms	Higher runtime (dual features + iterative expansion); parameters (circle size, etc.) need tuning
Chen et al. (2022b)	GLRT noise model	Simplified heteroscedastic noise model; LRT / GLRT; per-channel fusion	Splicing (detection)	Analytically controlled FAR; camera-noise “fingerprints”; no training needed	Mainly image-level; assumes model fits imaging / quantization pipeline
Dixit and Bag (2021)	CenSurE + LIPID	CenSurE keypoints; LIPID; k -d tree matching	Copy-Move (robust)	Robust to reflection, non-affine, blur, noise; very fast via optimized matching	Keypoint-only, may miss very smooth small areas; clustering + RANSAC adds complexity
Elaskily et al. (2021)	ConvLSTM + CNN hybrid	ConvLSTM $\rightarrow$ Conv / MaxPool stacks; GAP + Dense; unified input; no handcrafted features	Copy-Move (image-level; binary)	Very high accuracy at 100 epochs; low test time ($\sim 1.29 / 1.38/4.26$ s)	Needs large labeled data; overfitting risk; sensitive to #epochs; no localization map
Liu and Pun (2020)	Adaptive-SVD noise + VND	Adaptive SVD local-noise; Vicinity Noise Descriptor; SVM	Splicing (localization)	Handles multi-object splices; better precision vs. prior noise-difference methods	Sensitive to small segments; needs accurate noise separation in textured areas
Sekhar and Shankar (2020)	Object-wise BAG	Block Artifact Grids (JPEG DQ) per object; cosine dissimilarity	Splicing (localization; double-JPEG cue)	Object-focused maps; competitive vs. global BAG / DCT	Relies on double-JPEG cues & segmentation quality; less general for RAW / single-JPEG or non-JPEG edits
Li and Zhou (2018)	Hierarchical Matching	SIFT; two-level matching; iterative region growing	Copy-Move (improved accuracy)	More true matches; expansion gives complete coverage; high accuracy	Slower (multiple matching rounds); needs decent seeds; can over-grow into similar background
Wang et al. (2019)	SURF + PCET with superpixel-aided matching	MBS superpixels; SURF; PCET; improved NN; RANSAC + morphology (localization)	Copy-Move (pixel-level masks)	Strong on bright smooth regions & similar genuine areas; robust to rotation / scale / blur / JPEG / noise; competitive F-scores	Morphology / thresholds need tuning; runtime grows with large tampered areas; heavy rescale / interpolation is hard
Alberly et al. (2018)	SIFT + FCM (fast CMFD)	SIFT; fuzzy c -means keypoint clustering to speed matching	Copy-Move (image-level)	$\sim 15\%$ lower time vs. baseline SIFT (MCC-220; authors' 353 imgs); similar accuracy	Binary decision (no dense mask); keypoint scarcity in smooth areas
Mohammed et al. (2018)	Fusion: CMFD + Resampling	Dense-field CMFD (Cozzolino, 2016) + resampling (Radon / learning) via conditional fusion	General tampering (image-level)	+8 – 10% gain on NIST Nimble 2017 ($\sim 10k$ imgs); complementary cloning + resampling cues	Depends on selected modules; localization coarse / indirect
Korus and Huang (2016b)	Multi-scale PRNU fusion	PRNU correlation at multiple windows; fusion; content-aware / CRF; segmentation-guided variants	General tampering (localization)	Better small-forgery detection & shape fidelity vs. single-scale; broad tests / code available	Needs reliable PRNU / reference & careful params; drops on heavy compression or flat regions
Zampoglou et al. (2017)	Web-image splicing eval	Benchmarking under web conditions (recompression / rescale)	Splicing (localization)	Clear robustness picture under JPEG / reposts; reproducible framework	Many methods degrade with multi-recompression / rescaling
Chen et al. (2017a)	CRF / blur edges	CRF-based intensity / gradient stats at edges; CNN blur classifier; edge-focused localization	Splicing / Copy-Move (edge-level)	Distinguishes optical vs. post-edit blur; robust to depth / motion-blur confounders	Best near edges; weak on low-texture / edge-poor regions; tailored to blur / retouch cases
Bondi et al. (2017)	Camera-CNN + clustering	CNN camera-model descriptors on patches; iterative clustering to produce a tamper mask	Splicing (localization) via camera inconsistency	Localizes alien-camera regions without PRNU references; effective when devices are mixed	Weaker for copy-move or same-camera splices; needs CNN trained on camera models
Prasad and Ramkumar (2016)	SIFT / HOG / SURF comparison	SIFT, SURF, HOG; hybrids SIFT-HOG, SURF-HOG	Copy-Move (single & multiple)	Hybrids outperform single features	Classic keypoint limits (flat areas); no extra geometric normalization

As shown in Fig. 9, it frames traditional forgery analysis as one shared pipeline, pre-processing $\rightarrow$ feature extraction $\rightarrow$ matching / analysis $\rightarrow$ decision & localization, fed by two evidence streams, copy-move and splicing. The CMFD side targets self-duplication inside the same image; keypoint, dense field, and block designs are alternative feature-matching strategies chosen by the analyst to trade speed (keypoints, ap-

proximate search) against mask fidelity (dense / block), while geometric consensus steps (e.g., RANSAC-style filtering) enforce a single plausible transform for the cloned region. The splicing side aggregates four orthogonal evidence families that expose cross-source compositing: statistical residuals (texture / Markov cues), compression-resampling artifacts (double-JPEG, periodicities), sensor traces (PRNU, CFA / demosaic,

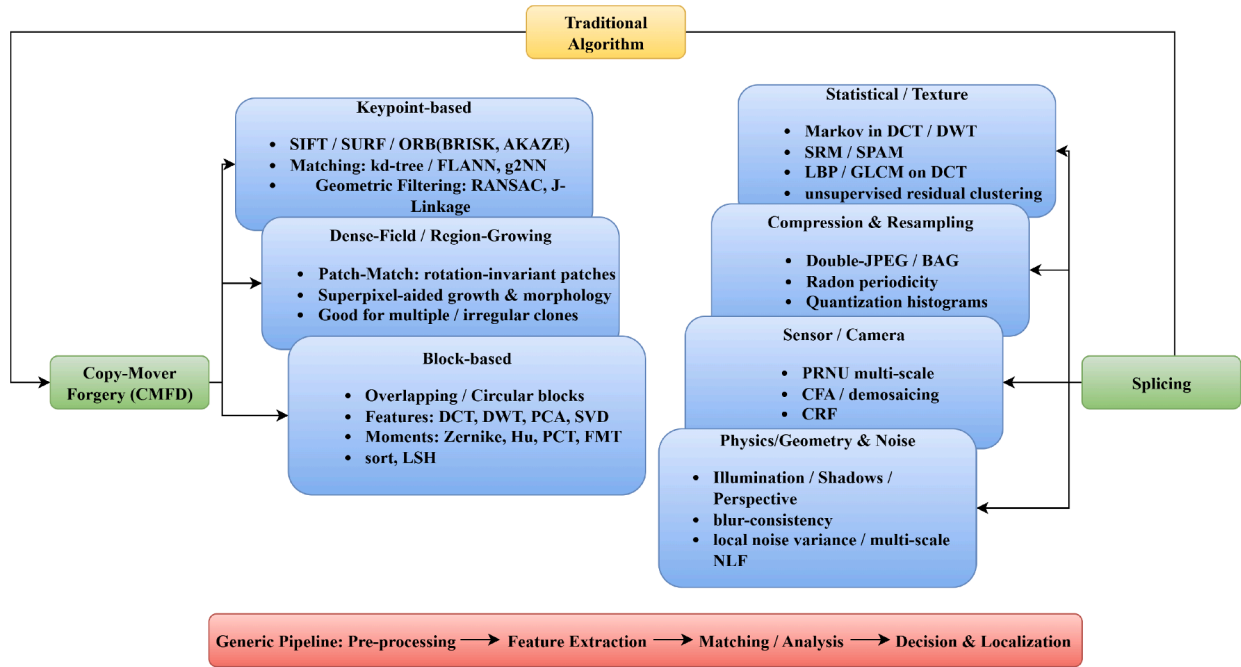


Fig. 9. Traditional methods used for copy-move and image splicing.

CRF) and physics / geometry inconsistencies (lighting, shadows, blur, noise scale). The bidirectional links imply that cues are complementary and can be fused across branches, e.g., a spliced paste may also carry resampling or blur inconsistencies that the pipeline can weigh alongside camera or statistical evidence. In essence, the diagram advises method selection by expected artifact and required granularity and advocates multi-cue fusion inside the same four pipeline stages for robustness to post-processing and varied attack strategies.

5.2. CNN-Based machine vision models

The rise of deep learning, particularly Convolutional Neural Networks (CNNs), revolutionized image forgery detection starting in the mid-2010s. CNN-based models automatically learn features from the data, which is a major advantage given the diverse and subtle artifacts left by modern forgeries. Instead of manually specifying what noise or edges to look for, a CNN can be trained on labeled examples of real vs. fake images and discover the statistical cues that differentiate them. Early attempts applied off-the-shelf CNN architectures (like AlexNet or VGG) to patch-wise classification: e.g., classifying small image blocks as either original or tampered with. However, it quickly became evident that standard CNNs, if trained naively, tend to learn the semantic content of an image rather than the forensic anomalies. For instance, if manipulated images in the training set contain certain objects more often, a CNN might latch onto those objects instead of learning manipulation traces. To address this, researchers devised specialized network layers and training strategies to focus CNNs on low-level anomalies while suppressing high-level content.

Modern image forensics has seen extensive use of Convolutional Neural Networks (CNNs) to detect and localize image forgeries. In the following, we curate key research articles (2015-2026) in four categories of forgery detection: (1) CNN-based copy-move forgery detection, (2) CNN-based image splicing detection, (3) detection of AI-generated (GAN-based) fake images, and (4) multi-stream & multi-task CNN architectures for localization.

5.2.1. Copy-move

Copy-move forgery involves copying a region of an image and pasting it elsewhere in the same image. CNN approaches have improved

detection by learning features that identify duplicated regions and distinguish source vs. pasted targets. Key methods include end-to-end networks for joint detection and localization, dual-branch architectures isolating similarity and manipulation traces, and two-stage frameworks combining deep matching with keypoint matching.

Liang et al. (2025) rethink copy-move detection as multitask learning and build MTFDN with hard-shared features plus two heads: global forged-homology detection and local manipulated-trace detection. Joint losses let the heads teach each other, yielding accurate localization and source / target discrimination on standard CMFD benchmarks. Al-haidery et al. (2023) design a hybrid pipeline that seeds candidates with SURF-HOG, grows them via SLIC superpixels, then confirms duplicates using overlapping Zernike-moment blocks. The method achieves high true positive rates with low false alarms and remains robust under geometric and photometric attacks. Kaur et al. (2022) fuse adaptive over-segmentation with Accelerated KAZE (AKAZE) and SIFT, operating in the Cr color channel to reveal subtle tampering artifacts. Experiments on MICC-F220, IMD, GRIP, and COVERAGE show improved precision / recall and resilience to rotation, scaling, JPEG compression, and noise, supported by ANOVA and cross-dataset tests. Diwan and Roy (2024) present a two-stage hybrid where CenSurE key points guide a CNN that learns data-driven features for copy-move detection and localization. The approach sustains accuracy across smooth, dense-texture scenes and withstands geometric transforms and common post-processing (JPEG, blur, noise, contrast / brightness changes). Wu et al. (2018) BusterNet is an end-to-end dual-branch CNN that fuses manipulation-trace cues with visual similarity cues and, uniquely, produces separate source and target masks. Trained with large-scale synthesis, it outperforms prior CMFD baselines on CASIA and CoMoFoD and remains robust to various attacks. Barni et al. (2020) tackle source-target disambiguation through a multi-branch CNN cast as hypothesis testing, learning features tied to interpolation artifacts and boundary inconsistencies. Designed to sit atop any CM detector, the tool reliably identifies the target region on synthetic and realistic data and improves on prior integrated solutions. Islam et al. (2020) propose DOA-GAN, whose generator uses dual-order attention derived from an affinity matrix to highlight copy-move locations (1st-order) and refine patch co-occurrence cues (2nd-order), while a discriminator sharpens masks. The network detects and localizes copy-moves

with explicit source / target labeling and achieves state-of-the-art results in challenging cases. Liu et al. (2021) build a two-stage framework: a self-deep matching backbone with atrous convolutions, skip matching and self-correlation-based spatial attention, followed by 'Proposal SuperGlue' that selects proposals and refines them via SuperPoint / SuperGlue matching. Bi and Pun (2018) presents a fast copy-move forgery detector that adapts coherency-sensitive hashing to find patch correspondences within a single image, then iteratively refines them using a new Local Bidirectional Coherency Error until the matches stabilize, at which point forged regions are read off directly.

5.2.2. Image splicing

Image splicing involves the composite images of different images. CNN-based splicing detection / localization methods often focus on detecting boundary artifacts, inconsistent noise or color filters, or anomalies in camera fingerprints. Many approaches use multi-task CNNs to output a pixel-wise mask of the tampered region. Below are key works:

Liu et al. (2022b) design PSCC-Net, a two-path architecture that extracts multi-scale features top-down and estimates masks bottom-up in a coarse-to-fine manner, with a spatio-channel correlation module to model spatial / channel dependencies. They report strong detection and localization with real-time throughput on 1080p images (50+ FPS). Zhu et al. (2024b) build an encoder-decoder tampering localizer that uses ConvNeXt blocks for hierarchical features and a UPerNet-style decoder to fuse them across scales. With combined losses and targeted augmentation, the method improves accuracy and robustness over current baselines. Zhang et al. (2023) revisit PRNU-based forensics and replaces handcrafted MRF fusion with a supervised, fully connected network that learns to combine multi-scale PRNU probability maps into reliable decisions. The approach reduces false positives and attains state-of-the-art pixel- and image-level accuracy, remaining resistant to noise and compression. Rao et al. (2021) introduce a multi-semantic CRF-based attention module that highlights boundary transition artifacts and re-weights CNN features at several depths. Coupled with ASPP for mask prediction, the system yields competitive or superior detection / localization across public datasets. Choudhary et al. (2024) combine Error Level Analysis with a VGG16-UNet to detect manipulation and segment altered regions. In CASIA-2.0, they report about 0.917 accuracy and 0.935 precision and provide a simple GUI for practical use. Li et al. (2023) propose ERMPC, a two-step scheme that first reconstructs edges via a context-enhanced graph and differentiable binarization, then controls regional message passing to decouple forged and authentic features. The edge-aware design sharpens boundaries and surpasses prior IFL methods on multiple benchmarks. Bi et al. (2019) present RRU-Net, an end-to-end splicing detector that augments U-Net with 'ringed' residual propagation and feedback to recall and consolidate discriminative cues. Improves localization without auxiliary pre / post-processing by better preserving contextual spatial information. Kwon et al. (2021b) develop CAT-Net, a dual-stream segmenter that learns forensic evidence jointly from RGB pixels and DCT-domain traces, with the DCT branch pre-trained for double-JPEG detection. Multi-resolution fusion delivers state-of-the-art splicing masks for both JPEG and non-JPEG images. Zhou et al. (2018) adapt Faster R-CNN to manipulation detection using two streams, an RGB branch for visual artifacts, and a noise branch seeded by SRM filters, with bilinear pooling to couple them at RoIs. The joint model outperforms single-stream and prior baselines in four datasets and remains robust to resizing and compression. Wang et al. (2022c) proposes a coarse-to-fine image splicing localization method that exploits inconsistencies in a camera's CFA interpolation. It first reconstructs the R/G/B channels by estimating interpolation coefficients with a blockwise covariance model and takes Laplacian residuals as 'forensic' features for coarse detection.

5.2.3. AI-Generated fakes (GAN)

The rise of GAN-generated fake content (e.g. deepfakes, synthetic faces) spurred numerous CNN-based detection methods. Early works

adapted image forensics to detect DeepFakes (face swaps) and GAN-synthesized images, often by spotting subtle artifacts left by generation (e.g. color inconsistencies, upsampling patterns). Later works incorporate frequency-domain analysis and attention mechanisms to generalize across generative models.

Liu et al. (2024d) introduce DA-HFNet, a progressive hierarchical detector-localizer that fuses RGB, noise, and frequency evidence through dual attention and multi-branch interactions across scales. They also released a GAN / diffusion dataset with whole-image and inpainting edits, and report sizable gains over recent baselines in both detection and mask quality. Ishrak et al. (2024) design a hybrid deepfake video pipeline that couples CNN and CapsuleNet with an LSTM head, then uses Grad-CAM to explain which facial regions triggered the decision. The goal is robustness to motion, rotation, and low quality while providing human-interpretable evidence for each classification. Rossler et al. (2019) present FaceForensics++, a large benchmark that covers Face2Face, FaceSwap, DeepFakes, and NeuralTextures under random compression and sizes. Supervised CNNs trained on these data markedly outperform human observers and remain reliable even with heavy compression. Guo et al. (2023) tackle unified image forgery detection and localization with HiFi-IFDL, representing each sample by hierarchical, fine-grained attribute labels and learning them with a multi-branch extractor plus dedicated detection and localization heads. The hierarchy, spanning 'fully synthesized vs. partially manipulated' down to specific methods, improves both image-level decisions and pixel-level masks across seven benchmarks. Fouad et al. (2020) target small splices in satellite imagery and propose two GAN / CNN variants: one with improved transfer learning in the discriminator and another with a re-designed CNN discriminator. In NASA / USGS data, small forgery detection increases from ~79% to 86% and then to 94%, while maintaining strong accuracy in medium / large forgeries. Huang et al. (2022) introduce FakeLocator, exploiting upsampling artifacts common to GAN generators to produce high-resolution grayscale 'fakeness' maps. Attention guidance, partial data augmentation, and single-sample clustering yield superior localization on FaceForensics++, DFFD, and multiple GAN sources, with resilience to JPEG, noise, blur, and low resolution. Xiao et al. (2023) reframe generalization as anomaly detection and train MCS-GAN solely on real faces; a Multi Criss-Cross Attention encoder plus a StyleGANv2 decoder reconstructs normals well but exposes fakes via reconstruction discrepancy. A latent-space encoding discriminator further sharpens discrimination, improving accuracy on Internet videos and public deepfake sets versus strong baselines. Xiao et al. (2023) eframes face-forgery detection as supervised anomaly detection by learning an artifact map that should be all-zeros for real faces and all-ones for fakes. A U-Net-style Artifact Map Detector (ResNet-18 encoder) is trained with triplet focal + metric-softmax losses to localize / amplify artifacts, while an auxiliary Xception classifier with circle loss makes features more discriminative; at test time, a simple artifact-map score (pixelwise average) replaces a binary classifier for better generalization to unseen manipulations.

Yang et al. (2024a) proposes a training framework that helps a detector keep its old knowledge while quickly adapting to new deepfake types, using multi-view knowledge distillation plus a hybrid replay sampler. Inside it, RSIA-Net fuses spatial textures and frequency cues with relation-aware interaction modules and a hierarchical spatial-frequency contrastive loss to extract fine-grained, generalizable clues, achieving superior cross-dataset performance on four public datasets. Zhang et al. (2025) combines a CNN backbone with a Mixture-of-Experts block and a Locally-Enhanced Vision Transformer (LEVT), and introduces Multi-Attention Scaling (MAS) to prevent attention collapse in deep transformers. A deepfake self-distillation (DSD) scheme auto-generates soft labels for better supervision, leading to state-of-the-art results across five deepfake datasets with strong robustness to shifts. Yu et al. (2024) tackle noisy ground-truth masks (from blending/compression) by explicitly modeling data uncertainty and supervising at three levels, pixel, patch, and image. This hierarchical and uncertainty-aware setup

improves generalization and delivers competitive performance on FF+++, Celeb-DF, and DFD. [Shang et al. \(2021\)](#) detects face forgeries by explicitly modeling intra-image inconsistencies: a Pixel-wise Relation (PR) module builds per-pixel relation maps and uses spatial attention to form “manipulated” vs ‘original’ region features, then a Region-wise Relation (RR) module compares those two region features with multiple metrics to decide if tampering exists, achieving SOTA in FaceForensics+++, Celeb-DF, and DFDC with robustness to image quality. [Zhu et al. \(2024a\)](#) decomposes two spaced video frames into illumination and reflection via a learnable frequency-domain Image Decomposition Module (IDM) trained in the log domain, swaps illuminations to recompose frames and amplify fake/real differences, and adds multi-level feature enhancement (FRRC + fusion blocks) to boost inter-frame inconsistency cues, outperforming prior deepfake detectors on mainstream datasets. [Liu et al. \(2024b\)](#) localize image manipulations with a selective-pyramid generator and dual homology-aware discriminators built on masked convolutions that judge whether pixels in predicted masked regions are homogeneous with the source, yielding state-of-the-art segmentation of tampered regions across four benchmarks. [Li et al. \(2025\)](#) frames deepfake detection as a domain-generalization (DG) task and builds a two-module detector: a shallow two-stream feature learner that extracts block-wise spatial and DCT-spectral patterns capture manipulation traces and source fingerprints, and a feature-based classifier that fuses them, applies mask-guided supervision to compare face vs. context, constructs robust correlation features, and corrects outputs using a confidence estimator.

5.2.4. Multi-stream and multi-task CNN

Advanced CNN architectures use multiple input streams or joint learning tasks to improve forgery localization. Multi-stream networks feed different modalities (e.g. RGB, noise, edges) into parallel CNN paths, and fuse them for richer features. Multi-task networks simultaneously learn classification (real vs. fake) and segmentation (tampered mask) or other auxiliary tasks, improving feature sharing and accuracy. Many state-of-the-art localizers adopt these designs:

[Lin et al. \(2023\)](#) introduce EMT-Net, a dual-trace architecture that extracts global / local noise with transformers and visual artifacts with CNNs, then enhances boundaries via edge artifact modules and explicit edge supervision. The approach strengthens localization under post-processing (e.g., smoothing, compression) and surpasses contemporary baselines on comprehensive metrics. [Chen et al. \(2017b\)](#) (DeepLab) establishes key building blocks for dense prediction: atrous convolution for controllable resolution, atrous spatial pyramid pooling (ASPP) for multi-scale context, and a fully connected CRF for sharp boundaries. These components form a strong segmentation backbone that many forgery localizers later adopt or adapt. [Yang et al. \(2023\)](#) propose DSSFN for hyperspectral image classification, a dual-stream self-attention network that fuses spectral and spatial signals and pairs them with a band-selection algorithm (SWGMP) to reduce redundancy. [Chen et al. \(2022a\)](#) design HFSRNet, an encoder-decoder that hybridizes LSTM-based patch traces with rotating residual features and applies semantic reinforcement in decoding to keep the same-level features consistent. The network handles copy-move, splicing, and removal, delivering state-of-the-art results on NIST16, COVERAGE, and CASIA. [Zhuang et al. \(2023\)](#) propose ReLoc, a restoration-assisted framework that first reconstructs a high-quality proxy of a distorted forgery to re-expose tampering cues, then localizes edits on the restored image. Alternating optimization with a forensics-aware loss markedly improves robustness to JPEG compression and transfers across localization heads. [Yang et al. \(2024b\)](#) introduces D-Net, a dual-encoder CNN that detects and localizes image splicing forgeries at the pixel level. An encoder learns forensic ‘fingerprints’ that differentiate tampered with from authentic regions, while a fixed encoder supplies structural cues via wavelet features; a hierarchical fusion connects them across layers. A spatial pyramid global feature extraction module widens the global context of the model, improving multiscale localization.

[Guo et al. \(2024b\)](#) present CLANet, a dual-stream (pixel / frequency) and multi-scale attention architecture trained with contrastive learning to separate forged from authentic patterns while a constrained loss mitigates overfitting. The design yields more general representations across diverse types of manipulation. [Das et al. \(2022\)](#) introduce GCA-Net, coupling an EfficientNet encoder and signal-aware (ELA / steganalysis) layers with a gated context-attention decoder to capture long-range dependencies while suppressing irrelevant low-level content. In CASIA, NIST, and IMD, it reduces false positives in pristine images and improves AUC by about 4–6% against strong baselines under realistic post-processing. [Munawar and Oussalah \(2025\)](#) introduces DSCL-Net, an explainable dual-stream attention model for image forgery detection and localization: it runs an RGB stream and an SRM-filtered ‘noise’ stream through twin ResNet-50 encoders, fuses them with multi-scale attention, and uses classification, segmentation, and a contrastive head to predict real/fake and highlight tampered pixels. LIME and SHAP are used to show which regions drive decisions, improving trustworthiness. [Sheng et al. \(2025\)](#) proposes SSR-IFL, a multi-scale attention framework for localizing image forgeries in stereo super-resolution (SR) images. It has two key parts: MSFE, which extracts multi-scale features using dilated self-attention to capture fine pixel-level correlations, and BPFF, which progressively fuses features in both directions to reduce SR reconstruction artifacts. [Yan et al. \(2024\)](#) develop DIF-Net, an encoder-decoder with an adaptive convolution pyramid (depth-wise strip kernels + learnable scale weights) and a dual-stream RGB / HSV intermediate-fusion module that aligns channel / spatial semantics. Produces high-quality masks with fewer parameters and improves CASIA2 results by +3.3% F1 and +2.4% AUC over prior methods. [Sushir et al. \(2024\)](#) present a blind detector that pre-processes with WE-CLAHE, extracts features through Hybrid DTT and VGGNet, prunes dimensions using improved horse-herd optimization, and classifies with a deep convolutional capsule autoencoder (DCCAE). An adaptive density-based fuzzy clustering step segment like-pixels, delivering 99.23% (CASIA V1), 98.75% (COVERAGE) and 98.07% (GRIP) accuracies.

[Tables 9, 10](#) and [Fig. 10](#) synthesizes recent learning-based and hybrid image-forgery methods alongside representative CNN baselines, revealing a clear convergence toward multi-branch, multi-scale designs. Across detectors, the strongest systems fuse complementary evidence, RGB with noise / frequency traces (e.g., PRNU / Noiseprint, SRM / DCT), edge or boundary cues, and coarse-to-fine decoders, often under joint image- and pixel-level supervision or contrastive losses; transformer or attention modules are used to aggregate long-range context, while CRF / edge heads sharpen boundaries. Copy-move and splicing are handled within unified encoders-decoders or dual-stream pipelines, with several works adding OSN-robust training or restoration-assisted pre-processing to counter compression and resampling. Reported gains concentrate on F1 / AUC and mask quality across common benchmarks (e.g., CASIA, NIST, COVERAGE, IMD), but limitations recur: sensitivity to heavy post-processing and domain shift, dependence on large synthetic training sets, computational / parameter overhead, and occasional weakness on small or texture-less edits.

5.3. Transformer-based machine vision models

Transformer models, which are based on self-attention mechanisms, have recently been applied to computer vision tasks with great success. In the context of image forensics, Vision Transformers (ViTs) and related architectures represent the latest wave of innovation (post-2020). The key feature of transformers is their ability to model long-range dependencies: each patch of the image can be compared to any other patch, allowing for a global view of the consistency of the image. This is particularly attractive for forgery detection and localization. For example, a spliced object in an image might be globally inconsistent with the rest of the scene (different color tone, resolution, etc.), which a transformer could catch by comparing all patches, whereas a CNN might focus only on local texture. Transformers also provide flexibility in

Table 9
Summary of CNN-based machine vision models.

Ref	Method	Technique	CNN Architecture	Key Results	Limitation
Diwan and Roy (2024)	CNN-Keypoint Two-Stage Hybrid	CenSurE keypoints + VGG16 CNN; fusion classifier for pixel-level masks	VGG16 backbone; two-channel fusion classifier	Strong F1 on seven datasets; evaluates nine CMFD attack types; pixel-level localization	F1 degrades under harsh post-processing/JPEG and heavy noise per robust plots.
Liu et al. (2024d)	DA-HFNet	Dual-Attention progressive fine-grained detection + localization; multi-modal fusion (RGB, noise, freq.)	Hierarchical multi-branch CNN with dual attention and edge-aware loss	Outperforms prior SOTA on its DA-HFNet dataset; supports whole-image & partial edits (GAN + diffusion)	Code / dataset “to be released”; limited cross-benchmark numbers in preprint
Guo et al. (2024b)	CLANet	Contrastive learning + dual-stream (pixel / frequency) + multi-scale attention; constrained loss to reduce overfit	Enhanced dual-path backbone + multi-scale & dual-stream attention + contrastive head	Improved generality / transferability across splicing / copy-move / removal	Preprint; compute-heavier; cross-benchmark generalization still evolving
Yan et al. (2024)	DIF-Net	Dual-stream intermediate fusion (RGB+HSV) with Adaptive Convolution Pyramid (depth-wise strip convs) and learnable fusion weights	Encoder-decoder with pyramid encoders per stream; intermediate channel / spatial consistency fusion	+ 3.3% F1 and + 2.4% AUC on CASIA2 vs. prior methods; high-quality masks with fewer parameters; SOTA on several datasets	Dual streams increase implementation complexity; results reported on four major datasets
Sushir et al. (2024)	Hybrid DCCAE + ADFC	WE-CLAHE preprocessing; Hybrid DTT + VGG features; Deep Conv Capsule Autoencoder (DCCAE) + Adaptive Density-based Fuzzy Clustering (ADFC) for segmentation	CNN feature extractor + capsule autoencoder; clustering-based localization	Accuracy: 99.23% (CASIA v1), 98.75% (COVERAGE), 98.07% (GRIP)	Non-end-to-end clustering post-processing; broader generalization not shown
Das et al. (2022)	GCA-Net	Gated Context Attention (global non-local attention + gating) to capture subtle discrepancies / reduce false positives	EfficientNet encoder with ELA + SRM + Bayar conv inputs; densely connected attention decoder (U-Net + + -style)	~4.7% AUC improvement over prior networks; fewer false positives on untampered images	Robustness beyond reported regimes not fully characterized
Liang et al. (2025)	MTFDN	Multi-task CMFD: global forged homology detection + local minimal tampered difference; joint loss	Shared backbone with two task heads (GFHD & LMTD)	Better robustness & localization than single-task CMFD	Public code / detailed dataset list not provided in excerpt
Zhang et al. (2023)	PRNU Multi-Scale Fusion	PRNU-guided multi-scale tamper probability fusion	FC layers fuse PRNU correlations across six scales	~ + 9% pixel-F1 and ~ + 8% image-F1 over prior PRNU; robust to JPEG / noise	Needs reliable PRNU; weak if PRNU suppressed / low-texture
Alhaidery et al. (2023)	Robust CMFD scheme	SLIC superpixels + SURF / HOG + Zernike / Legendre; kNN clustering; SVM-based	Traditional pipeline with SVM head	Higher TPR / lower FPR than several baselines on four CMFD datasets	Parameter-sensitive; varies with no. of superpixels / features
Lin et al. (2023)	EMT-Net	Enhance Multiple Tampering Traces: transformer noise branch + CNN RGB branch + edge-artifact enhancement	Transformer (global / local noise) + CNN (RGB) with edge & region decoding	Outperforms SOTA; robust under JPEG / smoothing / filtering	Heavier model (dual branches + transformers); more hyperparameters to tune
Yang et al. (2023)	DSSFN (HSI)	Dual-Stream Self-Attention Fusion for spectral-spatial modeling (HSI classification)	Dual streams (spectral + spatial) with self-attention; pyramidal residual conv; band-selection preproc	Higher OA / AA / Kappa than prior HSI classifiers on 4 datasets	Designed for HSI classification; applicability to generic IFDL is indirect
Chen et al. (2022a)	HFSRNet	Hybrid features + semantic reinforcement (LSTM on patches + rotating residual units)	Encoder-decoder with hybridization and semantic reinforcement in decoding	SOTA on NIST16, COVERAGE, CASIA among listed baselines	May miss tiny / thin forgeries; boundary-artifact reliance; tuning required
Zhuang et al. (2023)	ReLoc	Restoration-assisted localization: restoration module re-enhances traces; alternates training with localizer	Image restoration net + tampering localizer (e.g., FCN / UNet / MVSS-Net)	Significantly improves robustness to JPEG; restoration transferable across localizers	Sliding-window for high-res inference; alternate training adds complexity
Kaur et al. (2022)	AS + AKAZE + SIFT Hybrid	Adaptive segments + keypoints; Cr-channel pre-processing; multi-CM detection & aggregation	Hybrid traditional	Competitive accuracy & robustness across nine attack types; ablation on detectors / descriptors	Requires parameter tuning; block/keypoint trade-offs remain
Wu et al. (2022)	Robust IFD over OSN	Robust training with decoupled OSN noise: predictable + unseen	CNN detector trained with OSN noise modeling pipeline	Superior robustness vs. SOTA on OSN-degraded images; >5k-item OSN dataset; top ranking	Focus on OSN pipelines; robustness to non-OSN distortions not fully characterized
Wu et al. (2018)	BusterNet	Siamese FCN with copy-move branch + splicing/CL branch for joint localization/disambiguation	Dual-branch FCN; high-level semantics + low-level features; end-to-end	First deep end-to-end CM localization & disambiguation; competitive on USCISI	Coarse boundaries; poor localization/disambiguation on realistic data; identical labels to both regions

Table 10
Summary of CNN-based machine vision models with GitHub code.

Ref	Method	Technique	CNN Architecture	Key Results	Limitation	GitHub
Zhu et al. (2024b)	ConvNeXtFF	ConvNeXt encoder with multi-scale feature fusion for localization	ConvNeXt-B encoder + U-Net-style decoder; focal + Lovász loss	Best avg. F1 / IoU across 6 benchmarks; robust to common post-processing	Supervised mask training; some sensitivity to domain shift	https://github.com/multimediaFor/ConvNeXtFF
Dai et al. (2024)	CECL-Net	Complementary learning, Edge Extractor + Edge-Guided Feature Reconstruction, Pixel-wise Contrastive Supervision, Dense Fusion	Two complementary branches with edge extractor/reconstruction; multi-scale dense fusion	Outperforms 7 SOTA on 3 metrics across 2 benchmarks + 1 AI-manipulated + 2 real-challenge datasets	Higher training complexity; multi-branch + contrastive learning are compute-intensive	https://github.com/dream-sky1213/CECL-Net
Triaridis and Mezaris (2024)	MMFusion-IML	Early vs. late fusion of forensic filters on TruFor-style framework	Multi-modal fusion variants with dual-branch encoder / decoders	Competitive / SOTA via complementary forensic cues; code public	Fusion weights need careful tuning; extra compute vs. single-modal	https://github.com/IDT-ITI/MMFusion-IML
Guillaro et al. (2023)	TruFor	RGB + learned NoisePrint + + fingerprint; anomaly map + confidence map; global integrity score	Transformer-based fusion (SegFormer family) with anomaly & confidence decoders	Outperforms SOTA across cheapfakes + deepfakes; robust generalization; public code	Still subject to OOD manipulations; can vary with heavy laundering	https://grip-unina.github.io/TruFor/
Niloy et al. (2023)	CFL-Net	Contrastive feature-space learning to separate forged vs. pristine regions	CNN segmentation backbone + contrastive learning objective	Outperforms several existing methods on 3 benchmarks	Needs careful contrastive sampling/mining; can underperform on artifact-specific cases	https://github.com/Kishor-Bhaumik/CFLNet
Guo et al. (2023)	HiFi-IFDL	Hierarchical fine-grained IFDL (multi-label attributes) + joint detection / localization	Multi-branch extractor + localization head + detection head	SOTA across 7 benchmarks; unified over CNN-synth & image-editing domains	More complex training (multi-level attributes) than single-task methods	https://github.com/CHELSEA234/HiFi-IFDL
Liu et al. (2022b)	PSCC-Net	Progressive spatio-channel correlation; joint detection + localization for splicing / copy-move / removal	Top-down & bottom-up pyramid with spatio-channel correlation; lightweight detection head	Strong cross-dataset AUC / F1 on five different datasets	Generalization varies across datasets; relatively heavy model	https://github.com/proteus1991/PSCC-Net
Liu et al. (2021)	SelfDM + Proposal SuperGlue	SelfDM $\rightarrow$ score maps; DeepMask + SuperPoint/SuperGlue matching; ConvCRF refinement	VGG16 / ResNet50; proposals: DeepMask; keypoints: SuperPoint + SuperGlue	MICC-F600: best F1 at $\sim 832 \times 832$; CoMoFoD: two-stage boosts precision & robustness; MobileNetV3	Relies on good proposals; boundary FPs; look-alike confusions; high-res training unresolved	https://github.com/yaqiliu-cs/SelfDM-TIP
Barni et al. (2020)	Distool	Source / target disambiguation via 4-Twins Net + Siamese Net; score-level fusion	Multi-branch CNN (two specialized nets) with transform-based fusion	Outperforms BusterNet (except USCISD); robust to JPEG & local noise; end-to-end option	Needs accurate two-region masks; fails when CM localization can't isolate regions	https://github.com/andreacos/MultiBranch_CNNCopyMove_Disambiguation
Rao et al. (2021)	MS-CRF / HFSRNet family	CRF-based attention / hybrid features with semantic reinforcement	Encoder-decoder with attention + CRF-style refinement	Higher F1/AUC vs. earlier FCNs on NIST16 / CASIA; better boundary quality	May miss tiny / thin forgeries; boundary-artifact reliance	https://github.com/HighwayWu/ImageForensicsOSN
Chen et al. (2021)	MVSS-Net	Multi-View (noise / edge) learning + Multi-Scale Supervision (sensitivity / specificity balance)	Two-branch (noise-sensitive + edge-supervised) with multi-scale supervision; image & pixel heads	Generalizes across 5 benchmarks; competitive for pixel- & image-level detection	Two-branch training / inference cost; dataset-dependent in difficult cases	https://github.com/dong03/MVSS-Net
Kwon et al. (2021b)	CAT-Net	Compression Artifact Tracing with dual RGB + DCT streams	Two-stream HRNet / U-Net-like with RGB & DCT artifact volumes	SOTA mIoU on NC16 Splicing; strong on Columbia / Carvalho; robust at JPEG QF 60–90	DCT stream less useful for uncompressed images; re-compression can degrade	https://github.com/mjkwon2021/CAT-Net
Islam et al. (2020)	DOA-GAN	Dual-Order Attention GAN with ASPP and VGG-19 backbone for CM detection / localization	Generator with ASPP + Dual-Order Attention; Discriminator with ResBlocks; VGG-19 features	Improved F1 / accuracy vs. baselines (~ 6 – 12% F1 gains on USC-ISI / CoMoFoD)	Fails on uniform backgrounds & large-scale changes; sensitive to strong geometric transforms	https://github.com/ywgao123/DOA-GAN
Wu et al. (2019)	ManTra-Net	Unified manipulation-trace extraction + Local Anomaly Detection (LADN) for pixel-level localization	IMT extractor (IMC-ResNet / VGG) + fully-convolutional LADN decoder	Generalizes across manipulation types; robust to resizing / JPEG; ~ 0.8 s per 1024×768 image	Requires large synthetic training (IMC-385); not specialized for CM source / target disambiguation	https://github.com/ISICV/ManTraNet
Rossler et al. (2019)	FaceForensics	Benchmark dataset + baselines for facial manipulations (DF, Face2Face, FaceSwap, NeuralTextures)	Xception baseline and others	99.26% (raw) / 95.73% (HQ) / 81.00% (LQ)	Face-centric; not for generic splicing / copy-move on natural scenes	https://github.com/ondyari/FaceForensics
Zhou et al. (2018)	Two-Stream Faster R-CNN	Object-detection style: RGB + SRM-noise streams with bilinear pooling	Faster R-CNN (ResNet-101) with parallel RGB & noise branches	Competitive ACC / AUC / F1 on four different datasets; robust to resizing / JPEG	Box-level only (no pixel masks); degrades under severe compression	https://github.com/LarryJiang134/Image_manipulation_detection

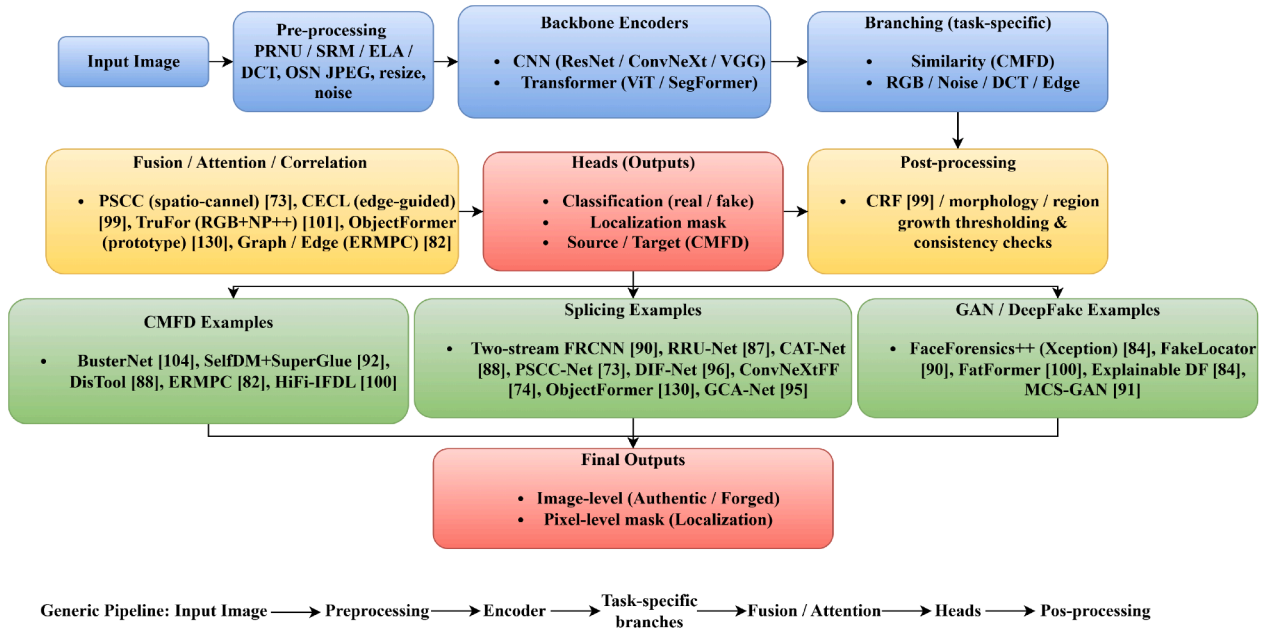


Fig. 10. CNN-based machine vision model architecture.

combining multiple input sources (patch embeddings from different modalities) into a unified analysis, which some recent works exploit for multi-signal forensics.

5.3.1. Vision-transformer (ViT)

The ViT's architectural design makes it particularly well-suited for image forgery detection and localization. Its self-attention mechanism, which models global relationships, is ideal for identifying the subtle spatially distributed inconsistencies that are a hallmark of sophisticated forgeries. For instance, a ViT can detect a mismatch in lighting or shadow across the image, a subtle yet critical cue that a local-focused CNN might miss.

The proposed pipeline for ViT-based forensics involves two main components:

- **Detection:** The model classifies an image authentic or forged. This is typically achieved with a classification head attached to the final output of the ViT encoder.
- **Localization:** To pinpoint the exact tampered regions, a segmentation head is added to the network. This head generates a detailed forgery map (heatmap) with a pixel-wise or patch-wise probability of forgery. This localization capability is crucial for forensic analysis, as it provides interpretable results and enables a deeper understanding of the manipulation.

Recent transformer-based advances for image forgery detection / localization (IFDL) coalesce around five themes: (i) adapting large pre-training, (ii) leveraging noise / frequency cues, (iii) feedback-driven coarse-to-fine decoding, (iv) hybrid generative / segmentation pipelines, and (v) robustness-oriented architectures. Liu et al. (2024a) aligns contrastive language-image pre-training (CLIP)-ViT features to text prompts using forgery-aware adapters and reports strong out-of-distribution detection (~98% on unseen GANs; ~95% on unseen diffusion), illustrating the benefit of language-guided adaptation, although it targets image-level decisions rather than masks. NFA-ViT couples a noise-guided dual-branch ViT with BR-Gen, a 150k localized AIGC dataset spanning object, stuff and background edits, producing state-of-the-art localized detection and improved transfer, but with added architectural complexity tied to the new data regime (Cai et al., 2025). A practical ViT + segment anything model (SAM) pipeline pairs transformer-based classification with fine-tuned segmentation for pixel masks across the CASIA, MICC, and

Columbia sets, although the results depend on SAM tuning and limited comparative reporting (Pawar et al., 2025). Hao et al. (2021) introduces dense self-attention encoders with correction modules to model all-pair interactions and delivers large gains over prior baselines on standard benchmarks. Zhu et al. (2023) injects feedback from a coarse map back into the early transformer layers and, with a contextual spatial pyramid plus a realistic MBH data generator, achieves leading generalization across nine datasets at the cost of a two-stage training pipeline. Complementary directions include (Atak & Yasar, 2024), which fuses reconstruction cues with ViT features for size-agnostic inference but offers sparse quantitative depth. Shi et al. (2025), which attains superior accuracy / robustness via a multi-exit ViT with cross-layer aggregation, graph reasoning, and cross-level enhancement; and Ravula and Singh (2025), which augments CMFD with GAN-generated forgeries and ViT features plus spectral clustering to reach ~99% on CoMoFoD, albeit with single-dataset bias. Finally, Li et al. (2024c) addresses boundary bias and pattern diversity through dynamic expert routing and importance-aware tokenization, improving mask quality and generalization, Liu et al. (2022a) increases cross-dataset face-forgery detection by fusing multi-level wavelet and spatial features within transformer blocks.

Li et al. (2024a) present CLIP-IFDL, a noise-assisted prompt-learning framework that builds on a frozen CLIP: instance-aware dual-stream prompt learning adjusts learnable positive / negative prompts via a prompt-adjustment network and a transformer decoder-based embedding-adjustment network, while a forgery-enhanced noise adapter injects noise-domain cues into selected ViT layers through cross-attention and zero-initialized adapters; a lightweight U-Net-style decoder, FiLM-modulated by the learned text features, outputs pixel-level masks. DeCLIP keeps CLIP frozen and decodes intermediate ViT features with a convolutional upsampler, achieving strong localization, particularly in latent diffusion inpainting, and better out-of-domain transfer by training on the harder LDM regime (Smeu et al., 2025). UnionFormer unifies RGB / noise evidence with object-consistency cues via a BSFI-Net encoder and a transformer head under multi-scale supervision, yielding a single representation that serves both image-level authenticity and mask prediction (Li et al., 2024b). MGQFormer replaces heavy decoders with a mask-guided, query-based transformer: The ground truth-derived guiding queries align the learnable queries through a dedicated loss, accelerating convergence, and improving the fidelity of the boundary shape (Zeng et al., 2024). GP-Net runs CNN and Transformer branches

in parallel and fuses them via a TcFusion module to balance fine-detail capture with global dependencies, reporting superior F1 / AUC in CASIA and NIST and strong resilience to JPEG compression, scaling, and Gaussian blur, while noting performance sensitivity in very small training sets (Peng et al., 2023).

5.3.2. Swin-transformer

The Swin-Transformer (Swin-T) was developed as a direct response to the computational inefficiencies and scalability issues of the original ViT. It was designed to maintain the powerful global modeling capability of Transformers while significantly reducing computational cost, making it a more practical solution for high-resolution images. The core innovation of the Swin-Transformer is 'shifted window-based' self-attention mechanism. Instead of computing attention across the entire image, Swin-T limits the self-attention calculation to a smaller, non over-lapping local window. This reduces the computational complexity to a linear relationship with respect to the number of patches, or $O(N)$, where N is the number of patches within a window. To ensure that information can flow between different windows and capture global context, the windows are shifted between successive layers. This mechanism effectively aggregates local information into a global context over multiple layers without the quadratic computational burden of the original ViT. Swin-T also introduces a hierarchical architecture, similar to CNNs, that allows it to process features at multiple scales. It progressively merges image patches in deeper layers, reducing spatial dimensions while increasing feature complexity. This is a crucial advantage for forgery localization, as tampering traces can manifest themselves at different scales, from fine-grained pixel artifacts to larger structural inconsistencies. By combining the hierarchical representation of CNNs with the global modeling power of Transformers, Swin-T offers a highly efficient and effective architectural solution.

Transformer-driven forensics increasingly fuse complementary cues, expand to new domains, and tighten pixel masks. Yang et al. (2025) introduce a Swin-based U-shaped encoder-decoder that runs RGB and frequency streams in parallel, extracting high-frequency evidence via an FFT→high-pass→IFFT module and merging it with a hierarchical attention fusion block to surpass previous IFL baselines. Thakur and Rohilla (2025) broaden the evidence space with multi-modal volumetric inputs (SRM residuals, DCT-Y high-pass, Laplacian), a Global-Context Swin Transformer encoder and an FPN decoder, producing state-of-the-art image- and pixel-level results on Columbia, Coverage, CASIA, NIST16, and IMD20. Complementing passive IFL, an optimal semi-fragile watermarking scheme pairs MERW-guided embedding with a Swin-generated watermark in a quaternion wavelet / SVD pipeline, reporting high imperceptibility (~ 65 dB PSNR; SSIM ~ 0.999) and robust tamper localization on color and HDR JPEG (Aberna & Agilandeewari, 2024). A triple stream Swin encoder-decoder learns high-level RGB cues, JPEG error-level disparities, and noise-residual artifacts, with middle / late fusion delivering 96.2% on CASIA-v2 and precise pixel probabilities (Agarwal et al., 2025). For video, Singh et al. (2025b) integrates a 3D Swin-B encoder-decoder for future-frame prediction with a ConvLSTM Siamese discriminator, localizing manipulations via attention maps and patching, and attaining 98.97% detection and 100% localization in diverse sets. Finally, a remote-sensing transformer fusion strategy frames global forgery detection as classification and ensembles Swin-V1 / V2 and Twins backbones with ImageNet-1K pretraining and a circular data-split, achieving competitive accuracy and a top-10 finish in the ISPRS 2024 contest (Zhao et al., 2024).

5.3.3. Fusion transformer

A growing consensus in the digital forensics community is that no single feature domain or architectural approach can provide a universally robust solution for all types of forgery. To achieve true generalization, a detection framework must be able to fuse complementary information from multiple sources. This has led to the development of multi-modal and hybrid approaches that combine handcrafted forensic

features, metadata, and deep features for a comprehensive understanding of an image's integrity.

A compelling example of this is the concept of dual-domain fusion, which is central to the DFST-UNet framework (Yang et al., 2025). The framework employs a dual-stream encoder to process the input image in both the spatial (RGB) domain and the frequency domain. This dual-domain approach is motivated by the fact that high-frequency forgery information, which often contains critical tampering clues like noise inconsistencies and JPEG compression artifacts, is difficult to capture from the RGB domain alone. Using a dedicated high-frequency feature extractor module (HFEM) in a separate stream, the model can fully expose forgery traces that would otherwise be missed (Liu et al., 2024e). The fusion of these two domains provides a richer and more robust feature representation, enabling the model to achieve superior performance, especially in terms of generalization and robustness to diverse manipulation techniques.

Karageorgiou et al. (2024) introduces an object mask-guided attention and a token-fusion transformer capable of ingesting an arbitrary set of forensic signals (RGB, noise, compression, etc.), with 'stream-drop' regularization and reported average F1 gains of 12.1% (detection) and 20.4% (localization) over prior art, while remaining extensible to new signals without retraining from scratch. Wang et al. (2022a) ObjectFormer couples RGB with high-frequency (DCT) evidence and uses learnable object prototypes to enforce object- and patch-level consistency, achieving state-of-the-art manipulation detection and masks in CASIA, Columbia, Coverage, NIST16, and IMD20. Wang et al. (2022b) introduce face forensics, M2TR detects artifacts across scales via a multi-modal, multi-scale transformer, an RGB branch plus a frequency branch joined by cross-modality fusion, and is validated alongside the new SR-DF deepfake dataset, surpassing contemporary baselines. Sun et al. (2022) focusing on boundary integrity, ET (Edge-enhanced Transformer) runs a two-branch edge-aware encoder and a self-attention enhancement module to improve both pixel and edge localization, reporting F1 scores of 0.6335 (CASIA v1.0), 0.6805 (CASIA v2.0), and 0.9440 (NC2016) and stronger resilience under common post-processing. Liu et al. (2023) TBFormer adopts dual transformer branches (RGB and noise) with an attention-aware hierarchical-feature fusion module and a transformer decoder for mask reconstruction, demonstrating consistent gains across public benchmarks.

Nam et al. (2025) unifies multi-spectral attention (via 2D-DCT bases) with multi-scale spatial attention in skip paths and adds an edge-aware, difficulty-guided transformer decoder, yielding superior generalization across benchmarks. Guo et al. (2024a) build a two-branch enhanced transformer in SegFormer, strengthen encoder features, and fuse RGB / noise streams with coordinate attention at multiple scales, achieving state-of-the-art robustness in standard datasets and their OSN variants. Liu et al. (2024c) adapt a frozen visual foundation model through prompt tuning and aligns its semantic tokens with BayarConv-derived high-frequency cues via a feature alignment and fusion module, improving performance in eight public sets and withstanding common post-processing. Complementing these transformers, HPUNet assigns hierarchical multi-level labels and couples multi-type features within a UNet augmented by soft-threshold attention and dense cross-scale interaction, delivering accuracy / F1 gains on AI-generated forgeries (Liu et al., 2024e). Collectively, these studies show that semantics-aware fusion and progressive, coarse-to-fine decoding raise localization fidelity and cross-domain resilience, while added architectural complexity and dataset sensitivity remain key deployment trade-offs.

Fig. 11 shows a canonical transformer pipeline for image forgery detection / localization. From the input image, optional pre-processing extracts forensic residuals (e.g., PRNU / SRM / ELA, DCT / JPEG grid, RGB / YCbCr, denoise / resize) before patchification and positional embedding. Features are encoded with a ViT / Swin / SegFormer or hybrid CNN-Transformer backbone and then routed to task-aware branches that assemble similarity tokens for copy-move, multi-stream evidence (RGB, noise / frequency, DCT / edge) and optional

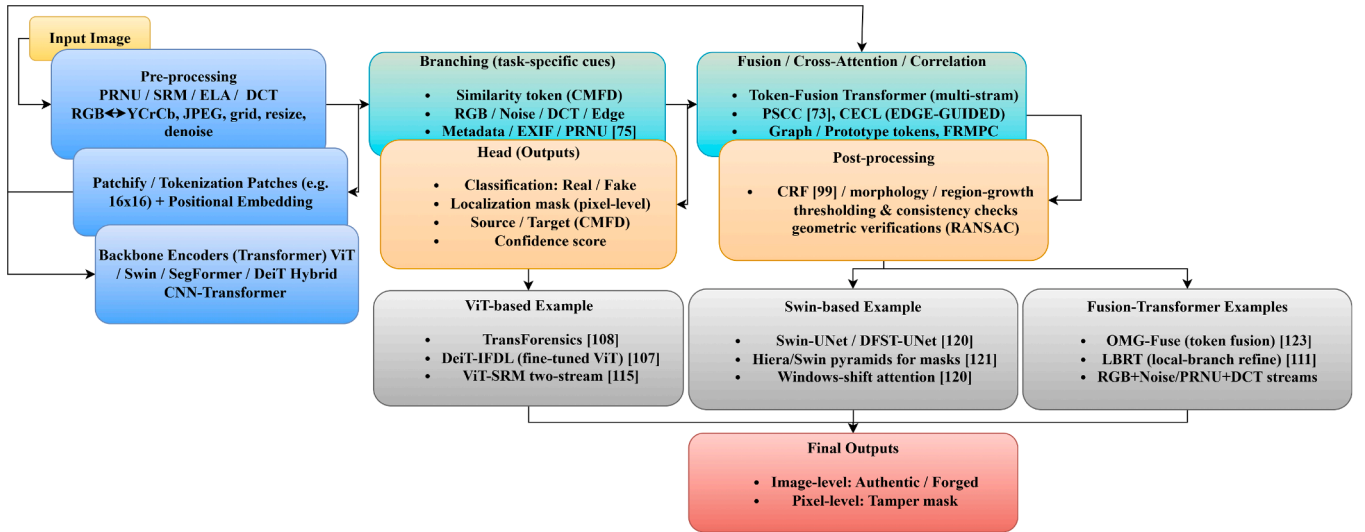


Fig. 11. Transformer-based machine vision model architecture.

metadata / PRNU cues. A fusion stage aligns these streams via token fusion, cross-attention, or correlation, after which lightweight heads output (i) image-level authenticity, (ii) pixel-level tamper masks, and, when applicable, (iii) source / target maps for CMFD; masks are refined through CRF / morphology / region growth and geometric consistency checks (e.g., RANSAC). Table 11 organizes representative transformer-era IFDL methods into ViT-based, Swin-based, and fusion architectures, highlighting recurring design patterns, noise / frequency-guided dual branches, SAM-assisted localization, global context Swin with multi-modal inputs, feedback-driven coarse-to-fine decoding, object-mask-guided token fusion, and CLIP-adapted encoders with prompt / adapter tuning. The table also records practical trade-offs: stronger cross-dataset performance and sharper boundaries often come at the cost of additional compute, sensitivity to pretraining/data distribution, or reliance on auxiliary modules (e.g., SAM) for precise masks.

6. Discussion

In this section, we discuss our key findings in light of the research questions that guide this review. The discussion synthesizes insights from the literature and reflects on the current state and future trajectory of image forgery detection and localization research.

RQ1 (Technique Categories and Evolution): What are the main categories of approaches to image forgery detection and localization developed between 2016 and 2025, and how have they evolved during this period? Our review finds that image forgery detection techniques can be broadly grouped into three generations: (i) Traditional methods based on hand-crafted features and classical machine learning, (ii) Deep learning methods primarily using CNN architectures, and (iii) Transformer-based or hybrid approaches leveraging self-attention mechanisms. Traditional methods relied on domain-specific visual artifacts, for example, sensor noise patterns, color filter array inconsistencies, resampling traces, or blockwise compression artifacts, combined with pattern matching or shallow classifiers (SVM, PCA clustering). These methods established the core forensic cues and are lightweight and interpretable, but often struggled with complex edits and had limited generalizability. Around 2016, the emergence of deep learning triggered a paradigm shift. Researchers began training CNNs to automatically learn forensic features from data, dramatically improving detection accuracy on benchmarks. CNN-based methods proved capable of capturing subtle anomalies that elude manual features, for instance, slight color tone mismatches or noise inconsistencies, and enabled end-to-end localization of manipulated regions (outputting pixel-level masks rather than just a binary decision). Early CNN models (e.g., AlexNet, VGG-

based) applied to patch classification were soon supplanted by architectures tailored for segmentation (Fully Convolutional Networks, U-Net variants) and multi-task learning (joint classification and mask prediction). By the late 2010s, nearly all new state-of-the-art methods were deep learning-based, often incorporating domain knowledge (e.g., custom layers to suppress semantic content and emphasize noise residuals). In the 2020s, there was another evolution with the adoption of vision transformers and hybrid CNN-Transformer models. These transformer-based approaches (e.g., ViT, Swin-Transformer backbones) can model long-range dependencies across the image, making them adept at catching inconsistencies that are spatially distributed (such as lighting or texture disparities between distant regions) which pure CNNs with local receptive fields might miss. Transformers have enabled detectors that consider the global context of an image, leading to more robust performance in cases like splicing (where distinguishing the spliced object from the background requires understanding the whole image). Additionally, many recent works fuse multiple streams of information, for example, combining an RGB image stream with a noise or frequency stream in a two-branch network (Karageorgiou et al., 2024). This fusion allows the model to concurrently analyze content and artifact cues.

RQ2 (Performance on Different Forgeries and Conditions): How do current forgery detection methods perform in different types of image manipulation (copy-move, splicing, inpainting, GAN-generated deepfakes), and what are their evaluation benchmarks and limitations? The surveyed literature shows that detection performance varies considerably by forgery type and the conditions under which detectors are tested. Traditional copy-move forgery detectors, for instance, are quite effective when the duplicated regions are large and the image has not been overly post-processed. Classic keypoint-based methods (using SIFT, SURF, etc.) can reliably spot duplicated features, and block-based correlation methods can catch even smooth-region clone stamp forgeries. For splicing (cut-and-paste from a different source), algorithms that look for inconsistencies in lighting, shadows, or camera-specific noise have reported high accuracies in curated datasets like CASIA (Liu & Pun, 2020). Many techniques achieve a detection accuracy well above 90% in within-dataset evaluations for copy-move or splicing forgeries. Similarly, for AI-synthesized fakes, such as deepfakes, CNN classifiers have achieved impressive performance on benchmark datasets. For example, the FaceForensics++ study showed that deep CNN detectors can exceed accuracy 95% in high-quality manipulated facial images, significantly outperforming human raters. However, these results often assume a consistent data distribution, i.e., training and test forgeries come from the same dataset or generation method.

Table 11
Summary of transformer-based IFDL methods.

Ref	Method	Transformer Type	Key Results	Limitation	GitHub
Cai et al. (2025)	NFA-ViT + BR-Gen	Noise-guided dual-branch ViT (forgery amplification via attention modulation)	Introduces BR-Gen (150k localized AIGC images) and achieves SOTA with strong cross-benchmark generalization; robust on small / low-contrast edits.	Added noise branch and task design increase complexity; trained around BR-Gen data distribution.	https://github.com/clpbc/BR-Gen
Pawar et al. (2025)	ViT + SAM	ViT for binary detection + fine-tuned SAM for localization	Evaluated on CASIA v1 / v2, MICC-F2000 / F600, Columbia; delivers practical classification plus mask localization pipeline.	Localization reliant on SAM fine-tuning; limited comparative numerics reported.	-
Yang et al. (2025)	DFST-UNet	Swin-Transformer U-Net with dual streams (RGB + frequency) and hierarchical attention fusion	Dual-domain encoder with an FFT→HPF→IFFT high-frequency extractor and Swin blocks improves localization / generalization; outperforms prior IFL on multiple benchmarks.	Added frequency branch and attention fusion increase complexity and compute.	-
Thakur and Rohilla (2025)	MDLFormer	Global-context Swin encoder + FPN decoder with multi-modal inputs (SRM, DCT-Y HPF, Laplacian)	State-of-the-art image / pixel-level results on Columbia, Coverage, CASIA, NIST16, and IMD20 via global context + low-level artifacts.	Multi-modal volumetric input and GCST encoder raise training / inference overhead.	-
Zhu et al. (2023)	ProFact	Feedback-enhanced SegFormer / Transformer (coarse→fine with CSPM)	Outperforms SOTA on nine public datasets; adds realistic MBH forged-sample generator and progressive two-stage training.	Two-branch feedback and staged training add implementation overhead.	https://github.com/multimediaFor/ProFact
Agarwal et al. (2025)	Triple-Stream Swin Encoder-Decoder	Three Swin streams (RGB, ELA, noise-residual) with middle / late fusion	Trained on synthetic composites; achieves 96.2% on CASIA-v2 with precise pixel probabilities by fusing high / low-level traces.	Synthetic training bias; ELA dependence ties performance to JPEG settings/resizing.	-
Thakur and Rohilla (2025)	MERW + Swin Semi-Fragile Watermarking	Swin-generated watermark within quaternion DT-CWT/SVD embedding (semi-fragile)	High imperceptibility (~65 dB PSNR; SSIM~0.999) and robust tamper localization on color and HDR JPEG images.	Watermarking (active) scheme; different threat model than passive IFDL; deployment depends on prior embedding.	-
Karageorgiou et al. (2024)	OMG-Fuser	Fusion transformer with object mask-guided attention; multi-stream forensic signals + token-fusion and long-range transformers	Exceeds prior art with +12.1% (detection F1) and +20.4% (localization F1) across seven datasets; robust and extensible to new signals.	Needs external instance-segmentation masks; multi-stream training adds complexity.	https://github.com/mever-team/omgfuser
Liu et al. (2024a)	FatFormer	CLIP-ViT with forgery-aware adapters and language-guided alignment	Tuned on ProGAN yet generalizes strongly: ~98% to unseen GANs and ~95% to unseen diffusion generators.	Framed for image-level detection; does not produce pixel masks.	-
Li et al. (2024c)	AdaIFL	Dynamic, importance-aware Transformer (expert routing + variable-length tokens)	Consistently outperforms prior IFL methods; adaptive token aggregation reduces edge bias and improves mask quality.	Dynamic routing / experts add engineering complexity and tuning.	https://github.com/LMIAPC/AdaIFL
Li et al. (2024a)	CLIP-IFDL	Frozen CLIP ViT with noise-assisted prompt learning (dual-stream prompts + forgery-enhanced noise adapters; transformer-decoder embedding adjustment)	Outperforms prior IFDL methods on accuracy / generalization while reducing false alarms; preserves open-world priors from CLIP.	Added prompt / adapter machinery increases complexity; primarily framed for detection with a lightweight mask decoder.	-
Smeu et al. (2025)	DeCLIP	Frozen CLIP ViT encoder + learned convolutional decoder for localization	Achieves high-accuracy localization on latent-diffusion inpainting and shows better cross-generator generalization when trained on LDM data.	Relies on CLIP features plus a sizeable decoder; evaluations centered on face / inpainting scenarios.	https://github.com/bit-ml/DeCLIP
Liu et al. (2023)	TBFormer	Two-branch transformer (RGB + noise) with attention-aware hierarchical-feature fusion and transformer decoder	Consistent gains for mask localization on public datasets; open dataset / models available.	Dual branches and fusion module increase compute/engineering cost.	https://github.com/free1dom1/TBFormer
Guo et al. (2024a)	EITLNet	Two-branch enhanced transformer (RGB + noise) with coordinate-attention fusion at multiple scales	Strong generalization and robustness on benchmarks and OSN-processed variants.	Depends on dual-stream training; results focus on listed datasets.	https://github.com/multimediaFor/EITLNet
Peng et al. (2023)	GP-Net	Two-stream CNN + Transformer in parallel with TcFusion for feature merging	Outperforms contemporary methods on CASIA / NIST and related sets; robust to JPEG, scaling, and blur via long-range modeling.	Sensitivity on very small training sets; added complexity from dual branches and fusion.	-
Liu et al. (2022a)	Multi-Scale Wavelet Transformer	Multi-scale wavelet-transformer with frequency-based spatial attention and cross-modality attention (DWT + unified transformer block)	Improves within- and cross-dataset face-forgery detection by fusing multi-level high-frequency and spatial cues.	Face-specific scope; not designed for general scene forgeries.	-
Hao et al. (2021)	TransForensics	Dense self-attention encoders with dense-correction modules	Models all-pairs context; reports clear gains over prior baselines on main benchmarks.	Dense attention can be compute-heavy; code not indicated.	-

RQ3 (Open Challenges and Future Directions): What are the main open challenges in image forgery detection and localization and what directions should future research pursue to address these challenges? Despite considerable progress, our survey underscores several unresolved challenges that present opportunities for further research. The foremost issue is the generalization and adaptability of forgery detectors. As noted above, models struggle with forgeries generated outside of their training distribution. This challenge will only grow as new image synthesis techniques emerge. An important research direction is developing domain-general or few-shot detectors that can quickly adapt to new manipulation styles, for example, using meta-learning or one-shot learning to recognize a novel fake with only a few examples, or designing features that are invariant to source-specific artifacts. Another challenge is robustness against counter-forensic measures. As detection algorithms improve, forgery tools evolve to produce more convincing fakes. We are already seeing GAN-generated images with minimal detectable artifacts. Additionally, a malicious actor can intentionally add false traces or adversarial perturbations to confuse the detectors. Current CNNs and transformers are known to be vulnerable to adversarial attacks in other domains, and this remains a concern for forensic reliability. Future work should explore adversarially robust models and ensemble approaches that make manipulation detection more resilient against an intelligent adversary.

For RQ1, achievements are the maturing taxonomy from traditional cues to CNNs to transformers, and the integration of device / physics features with learned tokens; remaining opportunity lies in principled fusion and interpretability (Farhan et al., 2024; Mehrjardi et al., 2023; Zanardelli et al., 2023). For RQ2, accuracy improved on canonical sets (CASIA, Columbia, IMD2020), and cross-dataset AUC / ACC rose with transformer fusion, yet robustness to unseen generators, compression, and wild conditions is still the main gap (Castillo Camacho & Wang, 2021; Dong et al., 2013; Li et al., 2024b,c; Ng et al., 2004; Novozamsky et al., 2020). For RQ3, open issues include fairness / geodiversity, dataset bias, and challenge-style generalization; your bars / lines can point to growing attention to these risks in 2023-2025 (Ju et al., 2024; Shankar et al., 2017; Torralba & Efros, 2011). To make the visual summary reproducible, we map each research question (RQ) to a 0–5 'Achievement' score and define 'Opportunities' as its complement (see Fig. 12).

RQ1 (Taxonomy & evolution): Achievement = $5 \times (\text{number of method families covered in review papers} \div \text{total families considered})$.

RQ2 (Performance & evaluation): Achievement = $5 \times (\text{median image-level accuracy across } \geq 3 \text{ public datasets} \div 100) + \text{bonus for localization support}$. Add +1 if pixel-level ground truth and a standardized evaluation protocol are reported (else + 0); cap the total at 5.

RQ3 (Open challenges): Achievement = $5 \times (\text{number of recent papers in our references that explicitly address explainability, robustness, or provenance} \div \text{total recent papers reviewed})$. Clip to [0,5].

For each RQ, **Opportunities** = 5 - achievements. All calculations use linear rescaling and are capped to remain within [0,5].

Economic costs: Fake images create both direct and indirect expenses: incident response and forensic analysis; platform moderation and take-down operations; brand/reputation remediation; legal counsel and settlements (e.g., defamation, IP, likeness misuse); fraud losses (insurance claims, investment scams, counterfeit evidence); and research pipeline disruption (paper retractions, duplicated experiments). A practical way to quantify the impact for a stakeholder S is

$$\text{Cost}_S = N_{\text{incidents}} (\text{analyst-hours} \times \text{rate}) + \text{legal} + \text{PR} + \text{lost-revenue}. \quad (1)$$

with sensitivity analysis over incident volume and OSN amplification.

Fake images impose costs across the entire response chain, analyst time, forensic tooling and moderation, refunds / charge-backs, legal / PR escalations, and longer-run brand trust and user churn, so we quantify exposure with a standard expected-loss framework from risk-assessment

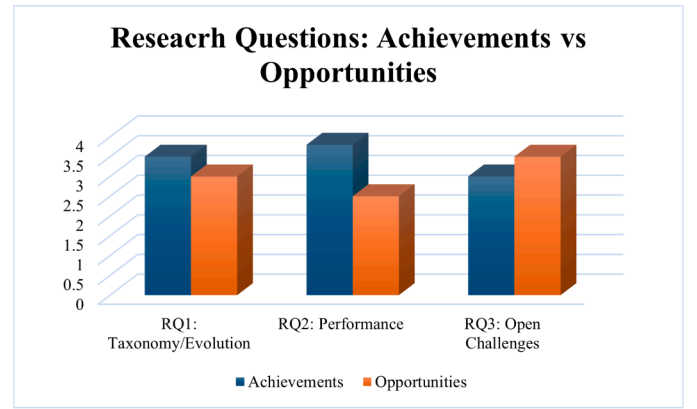


Fig. 12. Representation of achievements and opportunities of research questions.

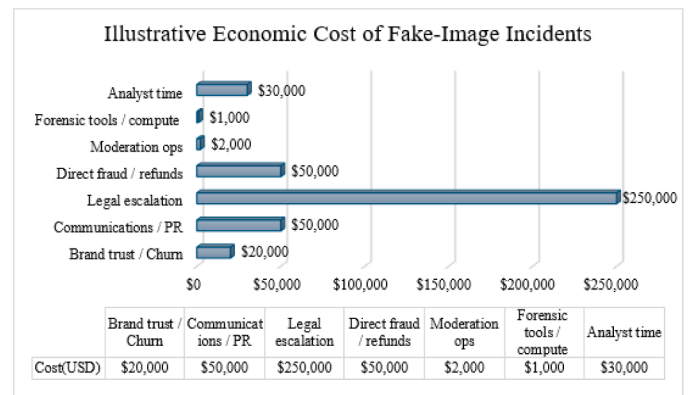


Fig. 13. Illustrative breakdown of the expected annual economic cost of fake-image incidents using an expected-loss model (per-incident components, probability-weighted legal / PR escalations, and brand-trust churn). Values shown are placeholders; replace with organization-specific data (Ross, 2012).

practice (sum of per-incident costs and probability-weighted escalations, plus a churn term), consistent with NIST SP 800-30 guidance on risk analysis (Ross, 2012). In research settings, the prevalence of manipulated figures is non-trivial, an audit of 20,621 articles found 3.8% contained problematic images, creating downstream costs when retractions occur; one study estimated a mean \$392,582 in direct funding wasted per NIH-funded retracted article (Bik et al., 2016). In industry, deepfake-enabled fraud has already produced material losses, from the \$243k AI-voice CEO scam reported in 2019 to the \$25m Hong Kong multi-participant video-call case in 2024, underscoring tangible financial risk that insurers now explicitly address in cyber / social-engineering coverage. Fig. 13 visualizes this cost model for an illustrative scenario (placeholders shown), making clear that legal/PR escalations and direct fraud/refund items typically dominate, while the brand-trust/churn term can exceed all others when user lifetime value or incident volume is high (organizations should substitute their own parameters).

Legal / policy violations: Image forgeries frequently violate privacy and personality rights (unauthorized likeness, deepfake face swaps), defamation, copyright / trademark, and data protection rules when biometric cues are processed without consent. In evidentiary contexts, altered media breaks chain-of-custody and authenticity requirements; platforms further treat this as a terms-of-service violation (coordinated inauthentic behavior, impersonation). Detectors should therefore report confidence, provenance cues, and limits of inference to support compliant decision-making.

Diversity & bias: Empirical studies show that media forensics and vision models can exhibit uneven error rates across demographics (race,

Table 12

Cross-dataset results of CNN-based models.

Ref	CASIA		NIST16		IMD20		Columbia		COVERAGE		WILD		DSO		CoMoFoD		DEFACTO		GRIP	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
ManTra-Net Wu et al. (2019)	81.7	-	79.5	-	-	-	82.4	-	81.9	-	-	-	-	-	-	-	-	-	-	-
OSN Wu et al. (2022)	83.3	-	76.4	-	-	-	88.3	-	-	-	-	-	82.3	-	-	-	-	-	-	-
MVSS-Net Chen et al. (2021)	75.3	83.9	73.7	-	-	-	70.3	98.0	82.4	73.1	-	-	-	-	-	-	57.2	57.3	-	-
EMT-Net Lin et al. (2023)	85.6	-	98.7	-	-	-	83.2	-	81.2	-	-	-	-	-	90.6	-	48.1	-	-	-
HFSRNet Chen et al. (2022a)	-	93.2	95.4	98.8	-	-	-	-	78.2	92.7	-	-	-	-	-	-	-	-	-	-
CLANet Zhuang et al. (2023)	90.6	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CFL-Net Niloy et al. (2023)	75.6	86.3	91.8	99.7	89.9	89.9	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CECL-Net Dai et al. (2024)	87.5	-	95.9	-	77.8	-	-	-	-	-	76.7	-	-	-	-	-	-	-	-	-
GCA-Net Das et al. (2022)	92.2	-	95.3	-	-	-	86.4	-	87.4	-	-	-	-	-	-	-	-	-	-	-
DIF-Net Yan et al. (2024)	94.6	-	99.7	-	86.8	-	99.2	-	-	-	-	-	-	-	-	-	-	-	-	-
WE-CLAHESushir et al. (2024)	-	99.2	-	-	-	-	-	-	98.7	-	-	-	-	-	-	-	-	-	98.0	-
TruForGuillaro et al. (2023)	95.7	81.3	87.7	66.2	-	-	94.7	98.4	92.5	77.0	-	-	93.0	-	-	-	-	-	-	-
PSCC-Net Liu et al. (2022b)	82.9	-	85.4	-	-	-	98.1	-	84.6	-	-	-	-	-	-	-	-	-	-	-
ConvNeXtFF Zhu et al. (2024b)	85.3	74.3	84.8	-	93.6	-	-	-	75.2	-	-	-	80.1	-	-	-	-	-	-	-
ERMPC Li et al. (2023)	87.6	89.5	85.6	-	96.8	-	94.4	-	-	-	-	-	-	-	-	-	-	-	-	-
RGB-N Zhou et al. (2018)	79.5	93.7	-	-	85.8	-	-	-	81.7	-	-	-	-	-	-	-	-	-	-	-

gender, age) and locales, which threatens fairness and reliability. For deepfake detection in particular, researchers have documented demographic disparities and proposed fairness-aware training to narrow gaps; popular press and academic reports both note higher false positive / negative rates for some groups (Ju et al., 2024). Beyond demographics, geodiversity matters: large vision datasets are often Amero / Euro-centric, and models trained on them underperform on images from under-represented regions; new geographically diverse benchmarks highlight these shifts (Shankar et al., 2017). These patterns reflect classic dataset bias and weak cross dataset generalization, so evaluations should report per-group metrics and test across distributions rather than only in distribution (Torralba & Efros, 2011).

Psychological perspective: Visual misinformation rides on well-known cognitive shortcuts. When content is processed fluently, it feels more plausible or 'true', which boosts acceptance; repetition further amplifies this via the illusory truth effect (Reber et al., 2004). On social networks, novel and high-arousal material (awe, anger, anxiety) spreads disproportionately, helping false news diffuse 'farther, faster, deeper' than truth (Vosoughi et al., 2018). Practical mitigations should therefore pair detector outputs with explanations and add friction that shifts attention to accuracy before sharing; randomized experiments show that brief accuracy prompts measurably reduce intentions to share false headlines and improve sharing quality (Pennycook et al., 2021).

Educational and social impact: Targeted media-literacy helps users resist manipulation: training in lateral reading and verification improves fact-checking performance, and prebunking / inoculation interventions build resilience to common deception tactics (Wineburg & McGrew, 2019). Beyond user education, institutions can complement passive forensics with content provenance, cryptographic capture attestations, and edit histories defined by the C2PA standard (e.g., Adobe's Content Credentials / CAI), and platforms are beginning to pilot such labels.

6.1. Gap analysis and key limitations

Despite significant progress, several gaps and challenges remain:

Generalization and Robustness: As noted, many detectors perform poorly on manipulations that differ from those seen in training. They are also vulnerable to simple counter-forensic measures. For example, adding a slight amount of noise or blur to a forged image can dramatically decrease the confidence of CNN detectors. This lack of robustness is problematic. This indicates that models may focus on brittle features (like a slight noise disparity) that can be easily masked by an attacker. Research in robust classification (e.g. adversarial training) is being adopted to forensics to address this. Some recent detectors incorporate adversarial examples during training to harden the model. Nevertheless, robustness to compression, resizing, noise, etc. is still not solved; many deep detectors see a big drop in accuracy if, say, the image is JPEG-compressed at quality 75 instead of 100 as they were trained on. *Gap:* There is a crucial need for detectors that maintain high accuracy under various image distortions and on unseen forgery methods.

Data Scarcity and Evaluation: Building large, diverse datasets of forgeries with ground truth remains challenge. Although datasets such as CASIA and Coverage exist, they are relatively small and contain somewhat naive forgeries by today's standards. Recent large datasets are oriented toward deepfakes (faces) or competition data that may not be fully public. The community lacks an ImageNet for forensics, a massive labeled dataset covering many object types, contexts, and subtle manipulations. As a result, supervised deep learning approaches risk overfitting to limited data. Moreover, evaluation is often done on the same distribution as training (which inflates performance). *Gap:* There is a need for more comprehensive benchmarking, for example, testing detectors on images collected from the internet. An important limitation is that ground truth for real-world fakes is hard to obtain (we might not know the exact tampered pixels in a fake meme circulating online). This

Table 13

Cross-dataset results of Transformer-based models.

Ref	CASIA		NIST16		IMD20		Columbia		COVERAGE		CocoGI		In-the-Wild		BR-Gen		DEFACTO	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
OMG-FuserKarageorgiou et al. (2024)	98.1	91.0	82.1	71.9	-	-	98.7	99.4	96.0	81.0	89.4	72.6	-	-	-	-	-	-
ObjectFormerWang et al. (2022a)	88.2	99.6	82.1	-	-	-	95.5	-	95.7	-	-	-	-	-	-	-	-	-
TBFormerLiu et al. (2023)	95.5	99.7	-	-	86.3	-	-	-	-	-	-	-	-	-	-	-	-	-
LBRTGuo et al. (2024a)	-	-	-	95.5	-	-	-	-	-	-	-	-	-	-	-	-	-	96.6
DFST-UNetYang et al. (2025)	-	-	84.5	85.5	-	-	95.6	-	83.7	-	-	-	-	-	-	-	-	-
MDLFormerThakur and Rohilla (2025)	-	-	91.2	-	88.0	-	95.9	-	91.6	-	-	-	-	-	-	-	-	-
ELAAgarwal et al. (2025)	96.0	96.0	95.0	-	-	-	87.0	-	-	-	-	-	-	-	-	-	-	-
NFA-ViT Cai et al. (2025)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	97.9	-	-	-
TransForensicsHao et al. (2021)	83.7	-	-	-	84.8	-	83.7	-	-	-	-	-	-	-	-	-	-	-
ViT-VAENetAtak and Yasar (2024)	-	-	-	-	86.4	-	94.2	-	94.2	-	-	-	-	92.3	-	-	-	-
MEAFFormerShi et al. (2025)	86.3	98.8	-	-	-	-	-	-	87.4	-	-	-	-	-	-	-	-	-
AdalFLLi et al. (2024c)	93.8	86.1	-	-	95.2	-	-	-	93.8	-	-	-	-	-	-	-	-	-
CLIPLi et al. (2024a)	92.5	89.7	-	-	97.8	-	97.6	-	94.3	-	-	-	-	-	-	-	-	-
UnionFormerLi et al. (2024b)	97.2	88.1	-	-	86.0	-	98.9	-	94.5	-	-	-	-	-	-	-	-	-
MGQFormerZeng et al. (2024)	88.6	86.2	88.3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GP-NetPeng et al. (2023)	88.4	99.7	-	-	-	-	-	-	94.5	-	-	-	-	-	-	-	-	-
SparseViTSu et al. (2025)	98.2	86.1	-	-	-	-	97.0	-	93.5	-	-	-	-	-	-	-	-	-
IML-ViTMa et al. (2023)	95.9	99.7	-	-	94.1	-	-	-	80.1	-	-	-	-	-	-	-	-	-

complicates the quantitative evaluation, and we often rely on anecdotal evidence.

Explainability and Trust: For forensic use in legal or journalistic settings, it is not enough to say 'this image is fake', one must show why. Traditional methods had an edge here: they could point out, for instance, 'the lighting vectors differ by 30 degrees between these two faces' or 'double JPEG artifacts are present in this region' and show the analysis. Current deep models mostly give an output without human-interpretable rationale. This is a major gap for adoption. If a neural network flags an image as fake, but we cannot articulate the evidence, it is hard to act on it. Some initial solutions (FakeShield etc.) attempt to generate explanations, but this area is nascent. *Gap:* Developing methods that provide forensic evidence along with decisions. Potential approaches include visualizing attention maps on input features or hybrid models that output human-readable cues.

Multiple Forgeries and Mixed Media: Many current techniques focus on a single tampering per image. However, an image can be tampered with in multiple ways (e.g., an object removed and another spliced in, plus cosmetic smoothing applied). Detectors often specialize (a copy-move detector might miss an additionally retouched background). Approaches like multi-task learning or iterative detection (detect one kind of forgery then remove it, and detect others) are not widely developed. Also, images often come as part of a video or multi-image set in real cases. Integrating information across frames (for video forgeries) or across images (like a set of related images in a story) is still a challenge, although multimodal research is focused on it. *Gap:* Handling complex, multi-step forgeries, and cross-image analysis in a unified framework.

Real-Time and Large-Scale Deployment: In scenarios like social network content moderation or surveillance, one might need to analyze

thousands of images per minute. Many cutting-edge detectors (with heavy CNN or transformer backbones) are too slow for such a scale without massive computation. In addition, they might produce too many false alarms to be useful in an automated pipeline. The precision vs. recall trade-off must often be tuned: e.g., a method might catch 95% of fakes but also wrongly flag 5% of genuine images, which could be untenable if those numbers translate to millions of false positives. *Gap:* More efficient algorithms and better calibration of detector confidence to operate at scale with minimal human review.

Practical deployment guidelines: Beyond algorithmic accuracy, real-world IFDL systems must satisfy strict constraints on latency, throughput, and memory. In large-scale platforms, it is rarely feasible to run a heavy transformer-based localizer on every image. A common engineering pattern is therefore to split the workload into (i) high-volume triage and (ii) low-volume forensic analysis. For triage, lightweight image-level detectors, such as shallow CNNs, pruned backbones, or models operating on downsampled images, can rapidly filter obviously benign content while keeping GPU memory and inference latency modest. Only the small subset of suspicious images is then forwarded to more expensive segmentation or hybrid CNN-transformer models that provide pixel-level masks or attention maps suitable for audit and evidence. From a systems perspective, it is useful to treat the IFDL component as a micro-service within a larger content-moderation pipeline. Upstream modules handle standardization (color space, resizing, compression normalization), while the IFDL service exposes a simple interface returning a confidence score and, where available, a localization mask. Decision thresholds can be tuned differently for risk-sensitive applications (where missed forgeries are unacceptable and higher false-positive rates are tolerated) versus user-facing feeds (where false alarms must remain very

low). Runtime and memory footprint can be controlled through batching, sharing a single backbone across multiple heads (classification, localization, tampering-type prediction), and applying compression techniques such as quantization or knowledge distillation to the architectures reviewed in Sections 5.2 and 5.3. Traditional feature-based detectors remain attractive for extremely resource-limited or legacy environments. CNN-based models typically offer the best balance between accuracy and cost for server-side deployment at scale, especially when combined with cascaded or multi-stage designs. Transformer and multi-scale hybrid models are most appropriate when maximum localization fidelity is required and sufficient GPU resources are available. We therefore argue that future IFDL work should routinely report latency, memory usage, and hardware configuration, so that practitioners can more transparently trade off algorithmic performance against deployment costs.

Forger Adaptation (Adversarial Aspect): It is often said that this is a cat-and-mouse game. Forgers will adapt their techniques to evade known detectors. Already we see methods like Neural Attention Field in some GANs that try to remove detectable artifacts. A worrying trend is that generative models can be trained with a loss that penalizes detection by a certain detector (essentially creating an adversarial example generator). The detector patterns could thus be 'memorized' and avoided by the generator. This adversarial scenario is well-known in cybersecurity (malware vs. antivirus) and will likely play out in image forensics. **Gap:** The field needs to explore generalizable features that are hard for forgers to avoid without ruining the image realism. In addition, developing detectors that can quickly adapt to new forgery techniques (perhaps through online learning or community-fed data) will be crucial.

In Tables 12 and 13, AUC is computed at the pixel level and ACC at the image level (authentic vs. tampered). Overall, CNN baselines achieve strong image, ACC on familiar sets (CASIA, Columbia, COVERAGE) but their pixel, AUC drops on harder, more diverse corpora (WILD, DEFACTO, DSO), showing sensitivity to distribution shift and post-processing; models that fuse forensic cues (noise / frequency / edges), e.g., MVSS-Net, EMT-Net, HFSRNet, TruFor, retain comparatively better localization, yet still exhibit a gap between high ACC and modest AUC. Transformer-era methods consistently lift both metrics across CASIA, NIST16, Columbia, and COVERAGE, with architectures that combine long-range self-attention and multi-stream evidence (RGB + noise / frequency / PRNU or DCT) and coarse-to-fine decoders, e.g., OMG-Fuser, TBFormer, DFST-UNet, MDLFormer, AdaIFL, delivering the strongest cross-dataset pixel-AUC while maintaining high image-ACC. The toughest regimes (In-the-Wild, BR-Gen, large synthetic composites) still depress scores, underscoring that robust generalization remains open; the evidence supports (i) multi-cue fusion plus global context as key to localization fidelity and (ii) the necessity of reporting both pixel-AUC and image-ACC rather than relying on classification alone.

7. Conclusion and future work

Digital image forgery detection has matured substantially over the past decade, transitioning from niche analytical techniques into a broad arsenal of data-driven methods capable of spotting ever more subtle manipulations. In this review, we reviewed more than 120 primary studies from 2016 to 2025 and presented a taxonomy that classifies existing techniques by their approach (traditional vs. CNN-based vs. transformer-based) and by the forgery type they target. We summarize key algorithms in each category, discuss commonly used datasets and benchmarks, and analyze how well current methods perform under various conditions. From this review, a clear narrative emerges: early forensic approaches laid the groundwork by identifying telltale pixel-level traces of tampering, and modern deep learning methods have built on those insights to achieve significantly higher accuracy and automation. Techniques such as CNN-based segmentation networks and attention-based transformers now dominate the state of the art, often achieving impressive results on standard evaluation sets. At the same time, our analysis

highlights that the field is still grappling with generalization and reliability issues; no single method is robust to all forms of forgery or all real-world perturbations.

The integrity of digital images is better protected today than it was a decade ago, thanks to advances in forgery detection. However, the work is far from complete. A recurrent theme in this paper is that *context matters*: a method tuned to a particular dataset or forgery artifact might excel in that narrow context but fail elsewhere. The community has recognized this, and there is a shift toward creating more generalized detectors and more challenging evaluation protocols. Based on our comprehensive review, we outline the following future research directions that could significantly enhance image forgery detection and localization in the coming years.

Cross-Domain Generalization: Developing techniques (e.g. domain adaptation, meta-learning) that allow models to detect manipulations in images from unseen sources or produced by new generative methods. Robust cross-domain features and training strategies will be crucial to handle the diversity of real-world media.

Adversarial Robustness: Investigating methods to defend against intentional evasion attempts. This includes making detectors resilient to adversarial perturbations and improving their ability to detect fakes that have been shielded by antiforensic operations (such as noise injection or purposeful blurring) to confuse algorithms.

Improved Localization and Attribution: Enhancing the granularity and accuracy of forgery localization. Future models should better delineate tampered boundaries and, for copy-move forgeries, determine which region is the source vs. clone. This might involve higher-resolution feature learning and combining visual cues with physics-based consistency checks (e.g., ensuring the lighting and perspective are coherent across an image).

Explainable Forensics: Integrating explainability into detection frameworks. As automated methods begin to be used in legal or journalistic settings, providing human-understandable explanations (such as visualizing detected noise anomalies or flagging inconsistencies in metadata) will increase trust and aid human experts in verification. Research into explainable AI-such as attention maps or prototype-learning approaches in forensics, is expected to grow.

Multi-Modal and Contextual Analysis: Extending image forensics to take advantage of additional context. The authenticity of an image might be verified not just through pixel analysis but also by cross-referencing with other information-for instance, matching the image against thumbnails from a video, checking consistency with a timestamp or GPS tag, or comparing with textual claims. Systems that combine visual analysis with contextual metadata or even crowd-sourced fact-checking could offer a more robust defense against misinformation.

Standardized Benchmarks and Open Challenges: The community would benefit from new public datasets that reflect evolving challenges-for example, datasets of 'difficult' forgeries that include multiple simultaneous edits, forgeries embedded in social media posts, or extremely realistic AI-generated scenes. Similarly, organizing regular evaluation challenges (on the model of the NIST challenges but open to all) can drive progress and foster collaboration.

By addressing these future directions, researchers can build on the strong foundation laid in the past decade. The continuing goal is to stay ahead of forgery techniques with detectors that are accurate, general, fast, and trustworthy. We hope that this survey not only encapsulates the current state-of-the-art in image forgery detection and localization but also sparks new ideas and efforts to tackle the next generation of image manipulation threats.

CRedit authorship contribution statement

Maryam Munawar: Conceptualization; Methodology; Data curation; Visualization; Writing - original draft; Writing - review & editing. **Mourad Oussalah:** Conceptualization; Supervision; Writing - review & editing;

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that there are no financial or personal relationships, equity holdings, paid consultancies, patent applications / registrations, or other interests that could be perceived to affect the objectivity of the research reported in this manuscript.

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Appendix A. Abbreviations

Below is a list of abbreviations and their full meanings as used in this paper for quick reference:

ACC	Classification Accuracy
ADFC	Adaptive Density-based Fuzzy Clustering
AI	Artificial Intelligence
AIGC	AI-Generated Content
AKAZE	Accelerated KAZE
ASPP	Atrous Spatial Pyramid Pooling
AUC	Area Under the ROC Curve
BAG	Block Artifact Grid
CASIA	Chinese Academy of Sciences Image Tampering Detection Evaluation
CMFD	Copy-Move Forgery Detection
CFA	Color Filter Array
CLIP	Contrastive Language-Image Pre-training
CNN	Convolutional Neural Network
CoMoFoD	Copy-Move Forgery Detection Dataset
CRF	Conditional Random Field
DCT	Discrete Cosine Transform
DeiT	Data-efficient Image Transformers
DF	Deepfake
DFST-UNet	Dual-Frequency Swin Transformer U-Net
DIF-Net	Dual-stream Intermediate-Fusion Network
DFDC	DeepFake Detection Challenge
DenseNet	Dense Convolutional Network
DWT	Discrete Wavelet Transform
EITLNet	Enhanced Image Tampering Localization Network
ECDC	Evolving Circular Domain Clustering
ELA	Error Level Analysis
EfficientNet	Efficient Neural Network
FLANN	Fast Library for Approximate Nearest Neighbors
FRCNN	Faster R-CNN
FMT	Fourier-Mellin Transform
FCM	Fuzzy C-Means
FCN	Fully Convolutional Network
FPN	Feature Pyramid Network
FPR	False Positive Rate
GLCM	Gray-Level Co-occurrence Matrix
GP-Net	CNN + Transformer parallel fusion network
g2NN	generalized 2-Nearest-Neighbor
GAN	Generative Adversarial Network
GLRT	Generalized Likelihood Ratio Test
HDR	High Dynamic Range
HOG	Histogram of Oriented Gradients
HPF	High-Pass Filter

IoU	Intersection over Union
JPEG	Joint Photographic Experts Group
LBRT	Local-Branch Refinement Transformer
LDM	Latent Diffusion Model
LRT	Likelihood Ratio Test
LSTM	Long Short-Term Memory
MISD	Multiple Image Splicing Dataset
ML	Machine Learning
MRF	Markov Random Field
NIST	National Institute of Standards and Technology
OSN	Online Social Network
ORB	Oriented FAST and Rotated BRIEF
PCET	Polar Complex Exponential Transform
PCA	Principal Component Analysis
PRNU	Photo-Response Non-Uniformity
RANSAC	Random Sample Consensus
RNN	Recurrent Neural Network
RQ	Research Question
RRU-Net	Ringed-Residual U-Net
RANSAC	Random Sample Consensus
ResNet	Residual Network
SAM	Segment Anything Model
SegFormer	Segmentation Transformer
SIFT	Scale-Invariant Feature Transform
SLIC	Simple Linear Iterative Clustering
SRM	Spatial Rich Model
SPAM	Subtractive Pixel Adjacency Matrix
SSIM	Structural Similarity Index
SURF	Speeded-Up Robust Features
SVD	Singular Value Decomposition
SVM	Support Vector Machine
Swin	Shifted Window
TPR	True Positive Rate
UADFV	University at Albany DeepFake Videos
VAE	Variational Autoencoder
VGG	Visual Geometry Group
ViT	Vision Transformer

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